

DESN2000 SOFTWARE ENGINEERING

Week5:
UX/UI and
all the wonderful things in-between

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Inspired by Dr Sasha Vassar's DESN2000 lecture slides (2022)

What are we doing today?

1. What is UI and UX
2. Why should we care about UI/UX?
3. Case studies
4. Some guidelines/things to consider
 - Guiding Principles of Good Interface

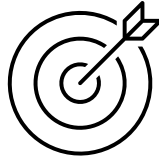
Where are we in the process?



Empathise

Brainstorming
User Research
Desk Research
Other techniques

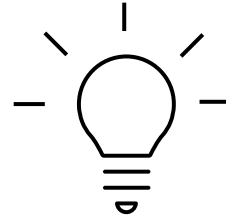
Personas
Scenarios
Journey Maps



Define

What is the problem?

Problem statements



Ideate

What are some solutions?

Themes that emerge from user research, categorise themes to understand initiatives. Think of your goals and features, which will set into the Epics and then break down each epic into user stories and associated acceptance criteria



Prototype

Start preliminary prototype

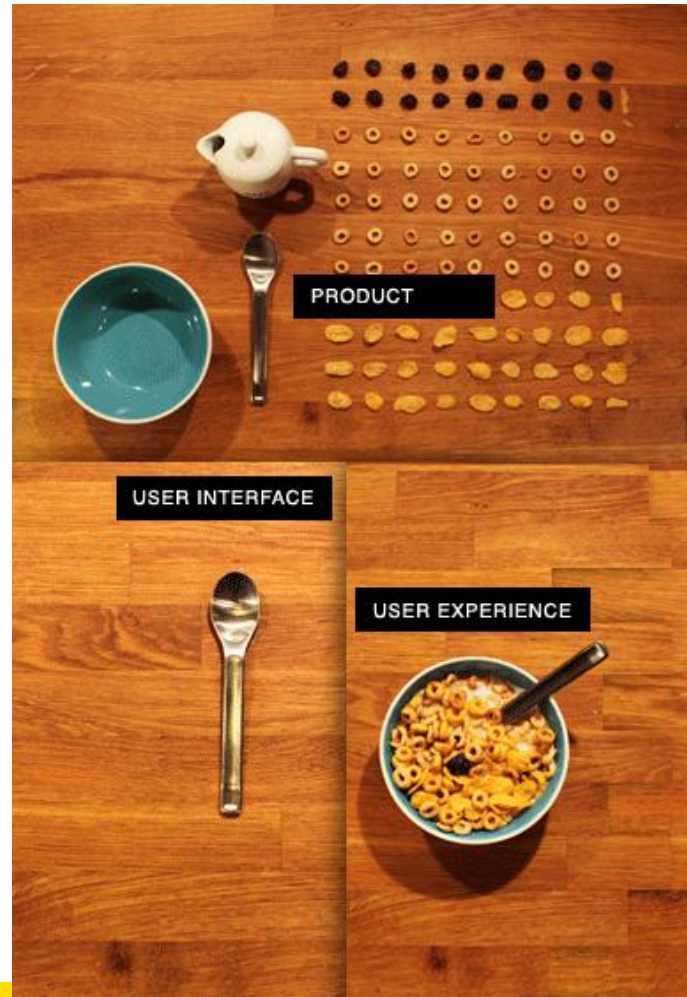


Test

What is UI/UX?

What is UI and UX

Two separate things!



<https://epiphycorp.com/cereal-ux/>

User Experience

"No product is an island. A product is more than the product. It is a cohesive, integrated set of experiences. Think through all of the stages of a product or service – from initial intentions through final reflections, from first usage to help, service, and maintenance. Make them all work together seamlessly."

-Don Norman

User Interface

Look and feel

- UI is the surface and the overall look of the interface
 - Screens
 - Buttons
 - Colours
 - Typography
 - Scrolls
 - Animations
 -
 - Other visual and interactive features

User Experience

Making an experience

UX is about **PEOPLE**

Creation of a user experience and everything it entails

- What problem am I solving?
- Who am I doing this for?
- How do people feel about the product?
- Does this align my product with its purpose?

User Experience

Making an experience

UX builds a positive (hopefully!) experience for the user, which helps to address business needs

Remember that even though you are building this product - you are not the user! So far in the process we have:

What is UI and UX

Two separate things!

UI
(User Interface)



UX
(User Experience)



Open top, shake
vigorously



Pray it doesn't get
all over you



Open top



Squeeze out
desired amount

What is UI and UX

For those of you that game



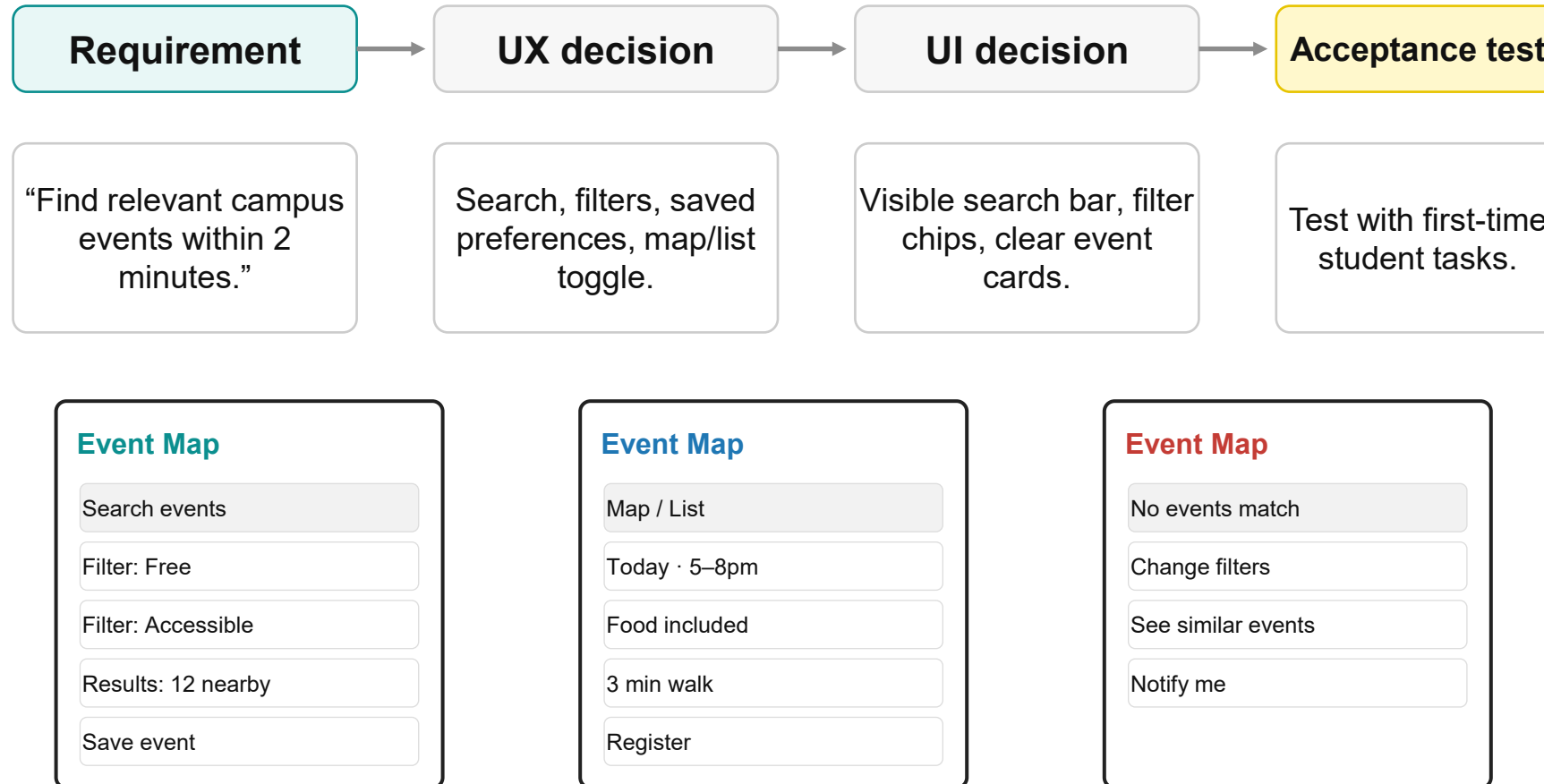
Image Credit:
https://static.wikia.nocookie.net/nintendo/images/6/62/Dinosaur_Land_-_Super_Mario_World.png/revision/latest?cb=20120224033322&path-prefix=en



Image Credit:
<https://www.wikihow.com/images/thumb/e/e0/The-Legend-of-Zelda-Link%27s-Awakening-save-screen.png/460px-The-Legend-of-Zelda-Link%27s-Awakening-save-screen.png>

Why should we care?

From requirement to interaction decision



Acceptance criteria are a UX contract

Acceptance criteria define the observable behaviours users should experience.

They force the team to design the “boring” but critical states: loading, empty, error, confirmation and recovery.

Functional behaviour

Given a student has opened the events page
When they apply “free” and “accessible” filters
Then the list shows only matching events 10 events per page.

Non-functional behaviour

Given the student applies a filter
When results update
Then results appear within 2 seconds and the page confirms how many were found.

Interaction consequence

Each criterion implies a screen state and microcopy:

- filter selected
- count displayed
- loading state
- empty-state recovery action

Example: evidence → requirement → interaction

Research observation
New students miss events because information is scattered across emails, Moodle and social channels.

Insight
The problem is not “no events”; it is poor discoverability and uncertainty about fit.

Requirement
Students shall discover relevant events using time, location, cost and accessibility filters.

User story

As a new student, I want to quickly find events that match my availability and comfort needs, so that I can participate without wasting time or feeling unsure.

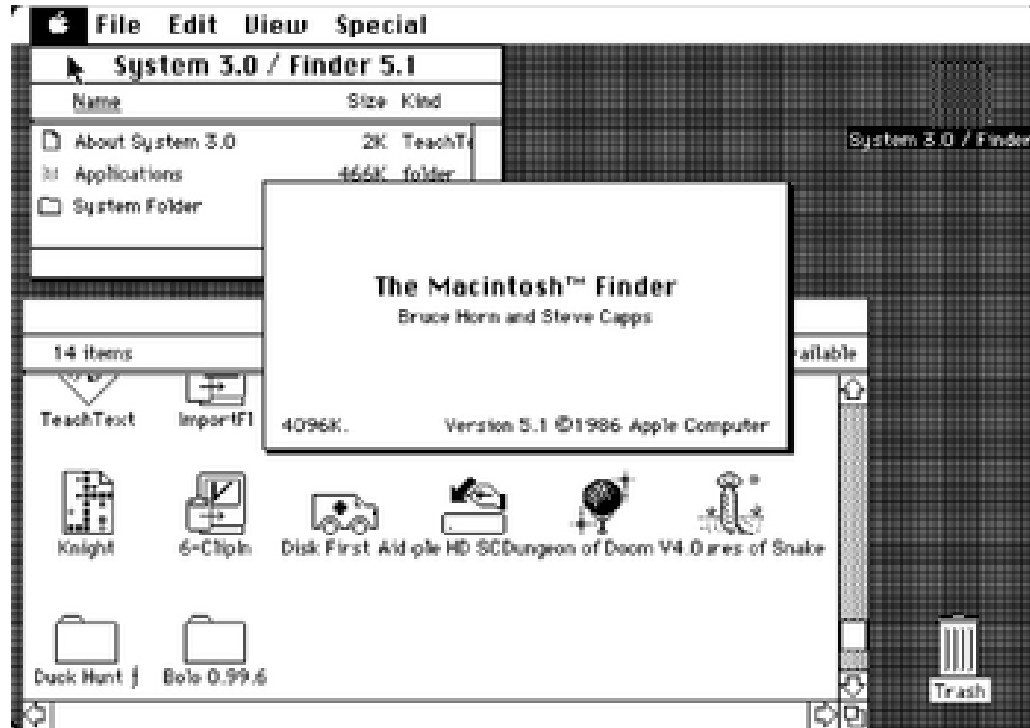
Design implication

The screen must reduce uncertainty: show location, time, cost, accessibility, crowd level and sign-up commitment before registration.

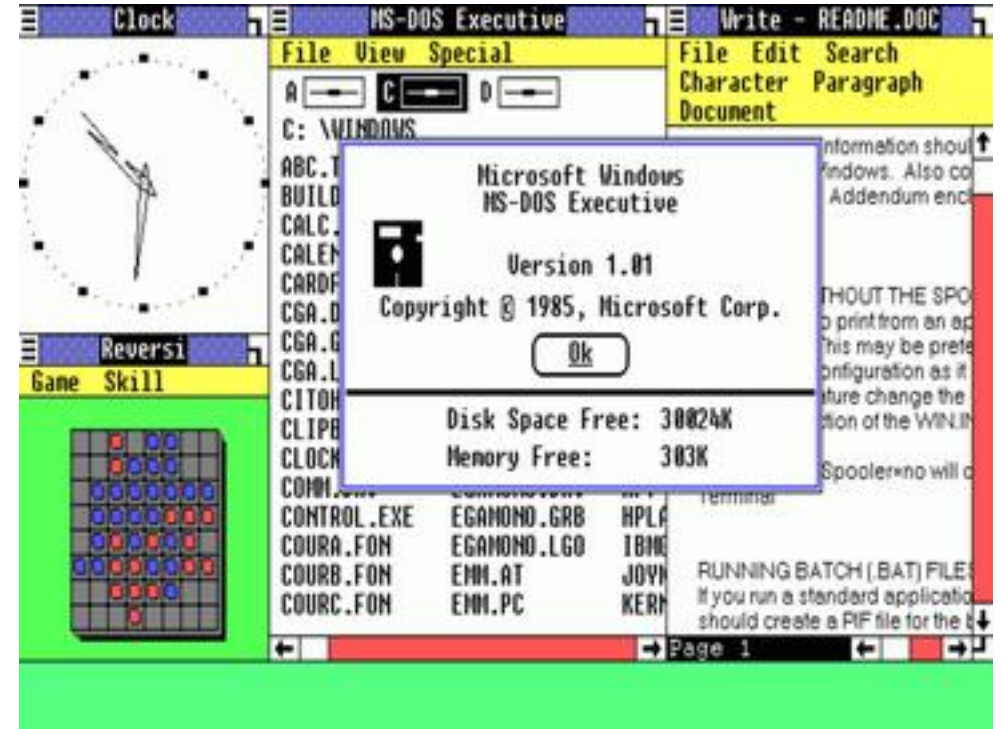
The interface is not invented from taste; it is justified by evidence.

Mac System vs Windows

1986



Mac System 3



Windows 1.01

https://apple.fandom.com/wiki/System_3

https://en.wikipedia.org/wiki/Windows_1.0x

Some experiences

UI

Please supply an image of your receipt to verify your claim

Receipt Number*

2646556

Receipt Date*

29/10/2019



UPLOAD IMAGE OF RECEIPT

File type : jpg

Size: 2.88 MB

File not uploaded. Please make sure your file is in jpg/png/gif/pdf format and less than 3.0 MB and then try again. You will not be able to continue until the file meets these requirements.

Retailer *

Harvey Norman

Retailer Suburb *

Moore Park

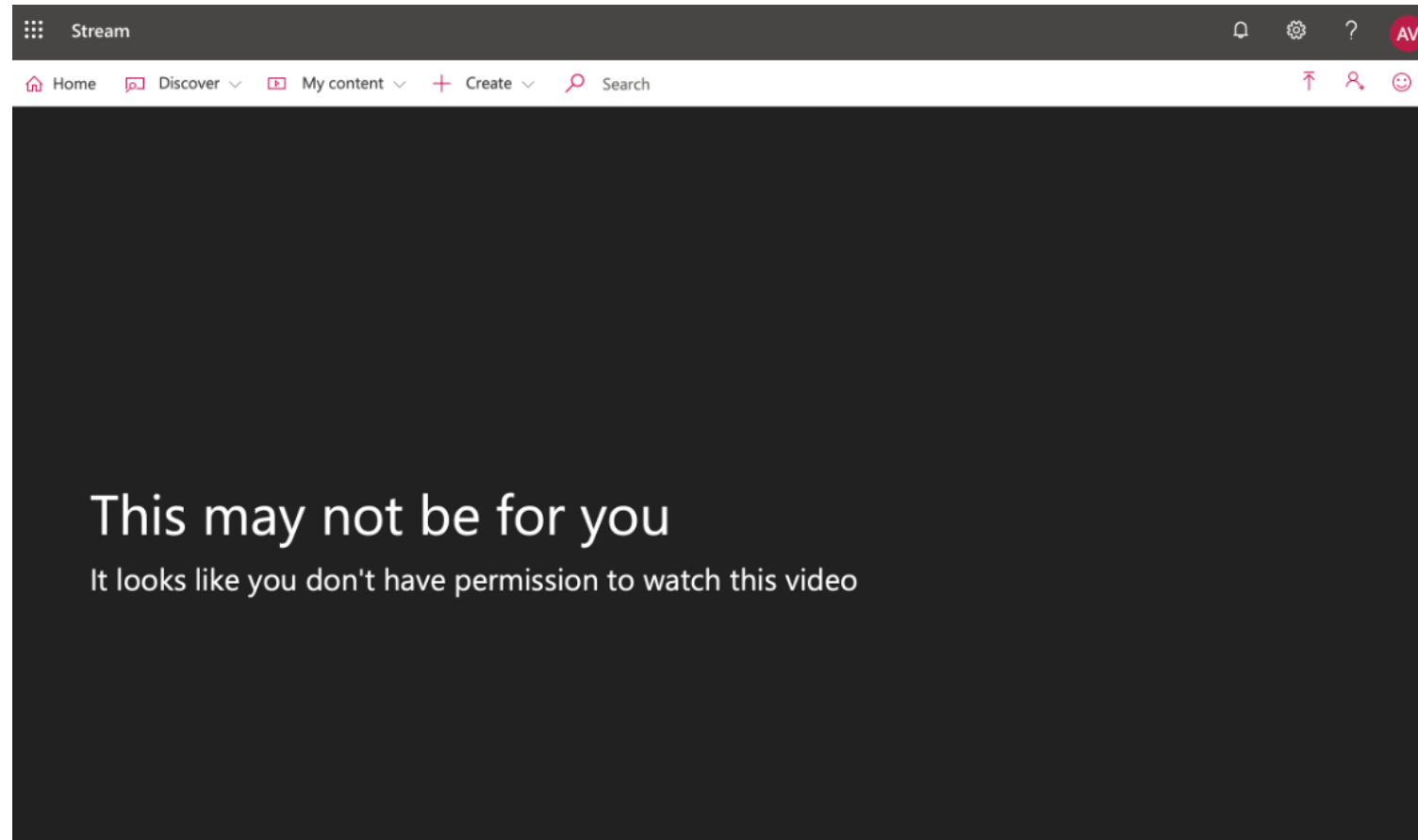
Retailer State *

NSW

ADD ANOTHER RECEIPT

Some experiences

UI



Why do we care about good UI?

UI

- Increased product use
- Increased sales
- Increased business loyalty
- Think about how you feel when you use a bad interface ...

Case Studies

Why do we care about UI and UX?

Why?

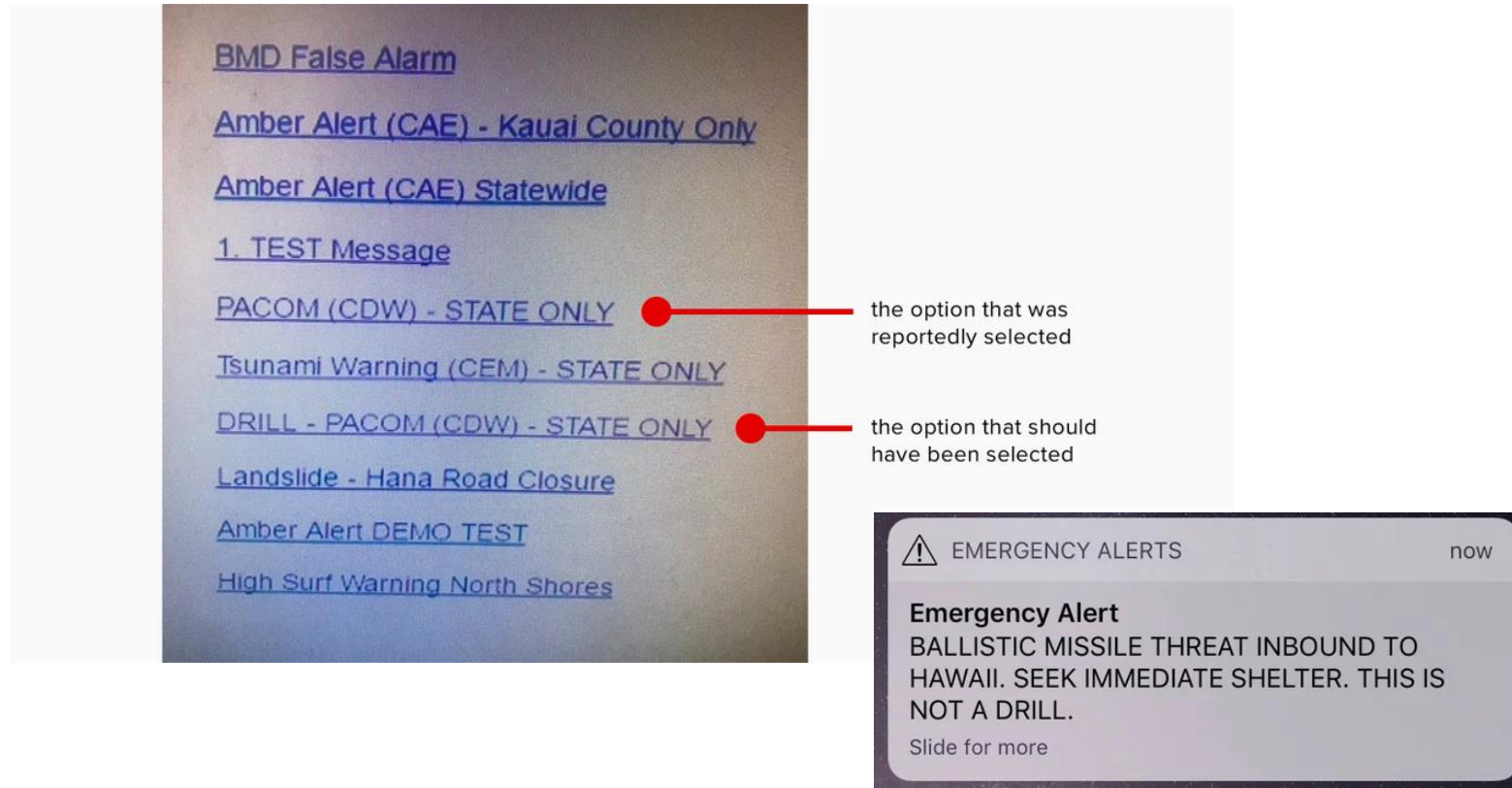


Image credit; <https://blog.prototypr.io/dangerous-drop-downs-%EF%B8%8Fbad-ui-design-caused-%EF%B8%8Fhawaiiis-false-missile-alert-5209cbc17f8c>

Why do we care about UI and UX?

Why?



Image credit: <https://www.pmp-corp.com/products/sti-flush-mount-push-button-emergency-stop-fuel-pump-shutdown-with-surface-cover/>

Why do we care about UI and UX?

Why?

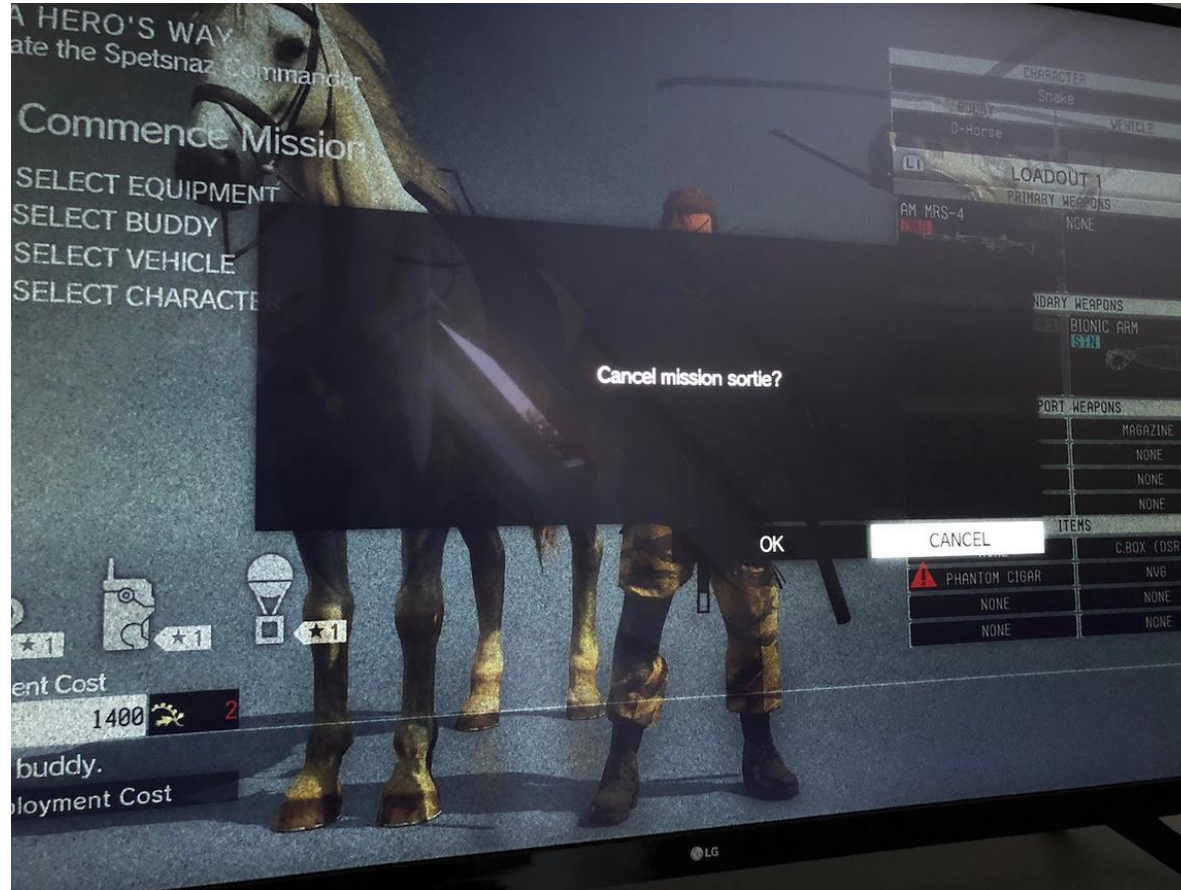


Image credit:

<https://medium.muz.li/weird-annoying-ux-patterns-in-triple-a-games-5c2cfe5c9aa9>

Guidelines

Guidelines

UI

- Easy to explore and easy to use
- User Control and Freedom
- Consistency
- Match between system and the real world Feedback
- Aesthetic and minimalist design Recognition rather than recall Error prevention
- Error diagnosis and recovery Help and Documentation
- **ACCESSIBILITY!**

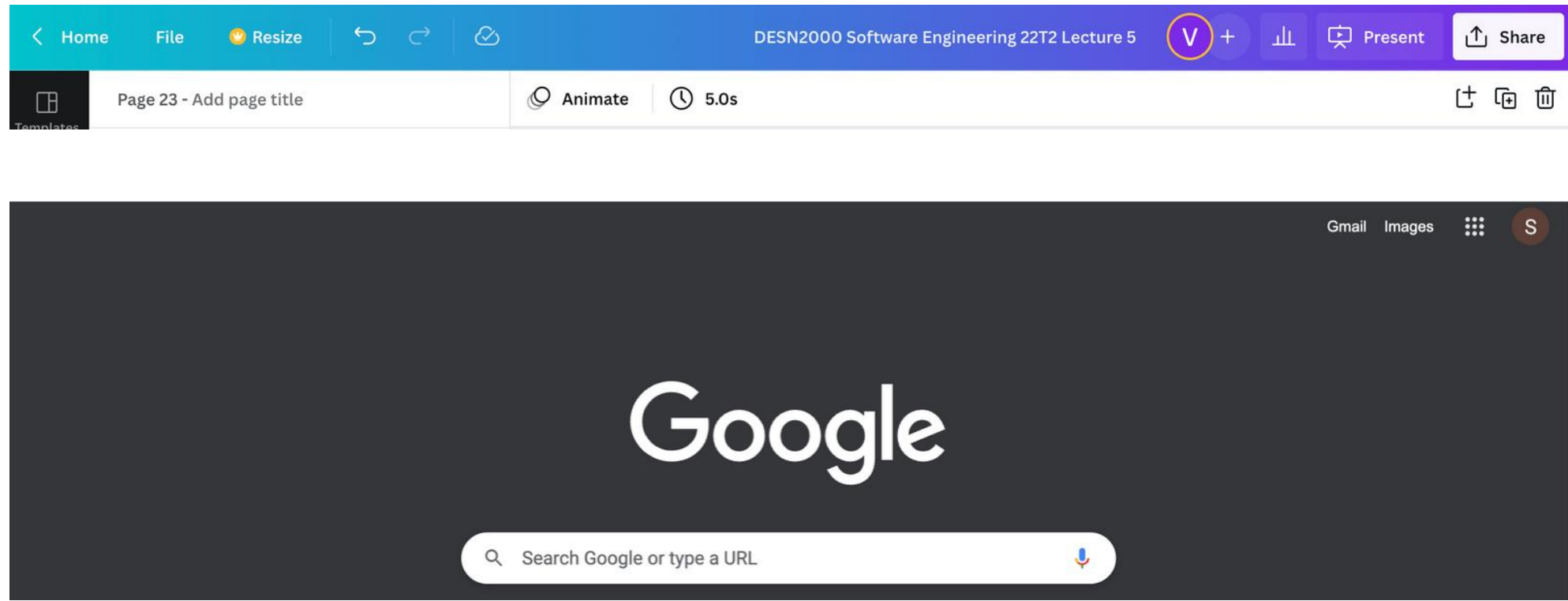
Easy to Explore and Easy to Use

UI

- Make the interface easy to explore and easy to understand
- Navigation should always be clear and self-evident.
- Put users in their comfort zone by providing some context of where they are, where they've been, and where they can go next
 - Visual cues
 - Predictability

Easy to Explore and Easy to Use

UI



Easy to Explore and Easy to Use

UI



Image credit: <https://www.polygon.com/god-of-war-guide-walkthrough-ps4-ps5/22264482/beginners-tips>

User Control and Freedom

- The user needs to feel that they are in control of the interaction
- Actions are not taking place in unpredictable, automatic ways
- It is obvious for the user to find an exit if they have traversed into an area of the application that they were not expecting

User Control and Freedom

The screenshot displays the Jira 'Default dashboard' interface. At the top, a navigation bar includes the Jira logo, a menu with 'Your work', 'Projects', 'Filters', 'Dashboards', 'People', and 'Apps', a 'Create' button, a search bar, and user profile icons. The dashboard itself is titled 'Default dashboard' and features four main widgets: 'Introduction', 'Assigned to Me', 'Projects', and 'Activity Stream'. Each widget has a title bar with icons for pinning, zooming, and refreshing. The 'Introduction' widget contains a 'Welcome to Jira' message with a link to 'Jira Software and Jira Service Management beginner guides' and a note about customizing text in the Administration section. The other three widgets ('Assigned to Me', 'Projects', and 'Activity Stream') are currently empty, showing only a small bar chart icon. The 'Assigned to Me' widget has a 'Refresh' button in its title bar. The 'Projects' and 'Activity Stream' widgets also have 'Refresh' buttons. The overall layout is clean and modern, emphasizing user control and freedom through customizable widgets and navigation options.

Default dashboard

Introduction

Welcome to Jira

Not sure where to start? Check out our [Jira Software](#) and [Jira Service Management](#) beginner guides or the [Atlassian training courses](#).

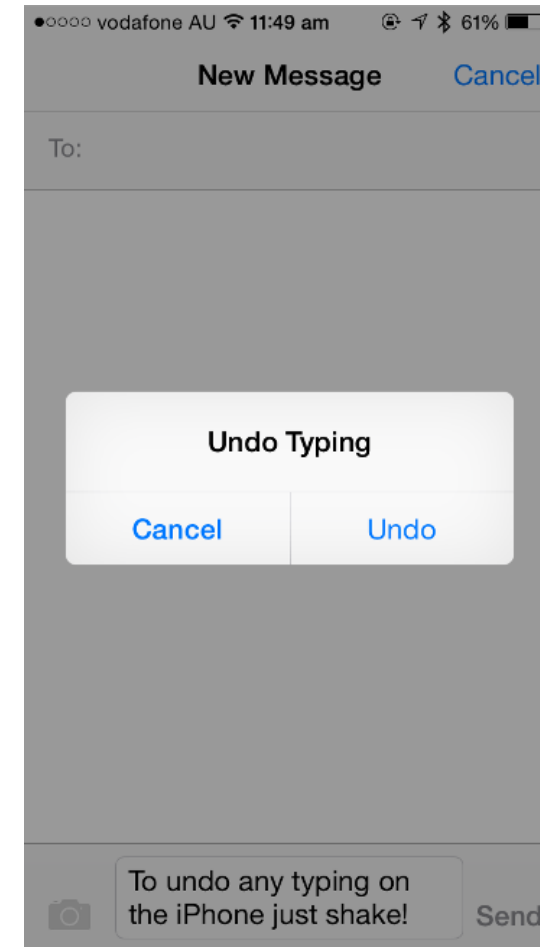
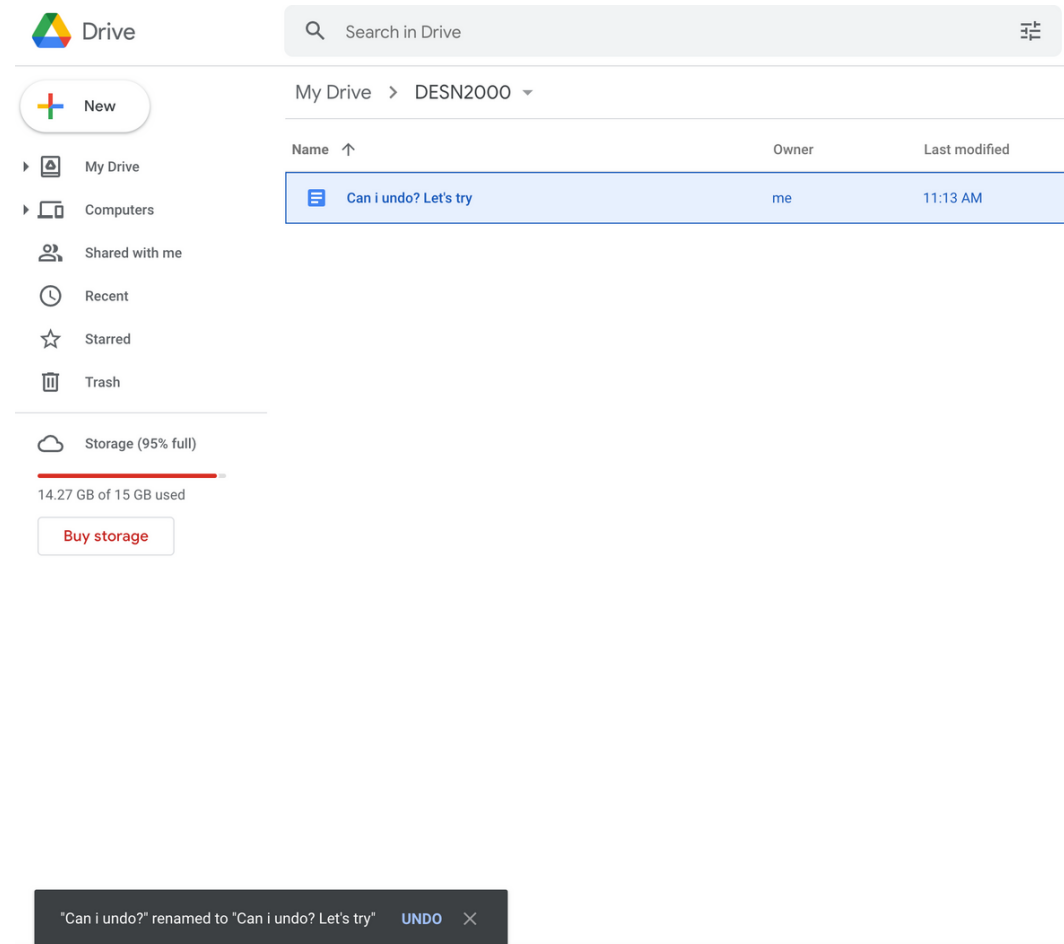
You can [customise this text](#) in the Administration section.

Assigned to Me

Projects

Activity Stream

User Control and Freedom



User Control and Freedom

The image displays two travel booking interfaces side-by-side. The left interface is Airbnb, featuring a search bar with 'Anywhere', 'Any week', and '2 guests'. Below the search bar are various filters like 'Lakefront', 'Beaches', 'Amazing pools', etc. The main content shows three property listings in Australia: Toukley, Coal Point, and Budgewoi. Each listing includes a photo, location, distance, dates, and total price. The right interface is Stayz, showing search results for 'New Forest National Park, England'. It includes a map with numerous property pins, a list of properties with details like '2 bedroom accommodation in Martin, Fordingbridge' and 'Studio Barn In The New Forest', and a sidebar with filters and a 'List your Property' button.

Airbnb Interface:

- Search bar: Anywhere | Any week | 2 guests
- Filters: Lakefront, Beaches, Amazing pools, Islands, National parks, Cabins, OMG!, The Arctic, Tiny homes, Design, Camping, Filters
- Property Listings:

 - Toukley, Australia**
74 kilometres away
25 – 30 July
\$5,079 AUD total
 - Coal Point, Australia**
99 kilometres away
7 – 12 July
\$4,896 AUD total
 - Budgewoi, Australia**
79 kilometres away
13 – 19 July
\$908 AUD total

Stayz Interface:

- Search bar: Where: New Forest National Park, Englan... | Check-in | Check-out | Guests
- Filters: Popular, Price, Rooms and spaces
- Property Listings:

 - 2 bedroom accommodation in Martin, Fordingbridge**
Sleeps 4 · 2 bedrooms · 3 beds
Excellent! 4.7/5 (9 reviews)
AU\$190 avg/night
 - Studio Barn In The New Forest**
Sleeps 4 · 1 bedroom
Exceptional! 5/5 (201 reviews)
Free cancellation up to 60 days before check-in
AU\$159 avg/night

Consistency

- Consistency helps users understand what they could do since they can build on knowledge they have acquired elsewhere (internal or external)
- Terminology means the same thing
- Actions or sequences of actions are conducted in the same manner

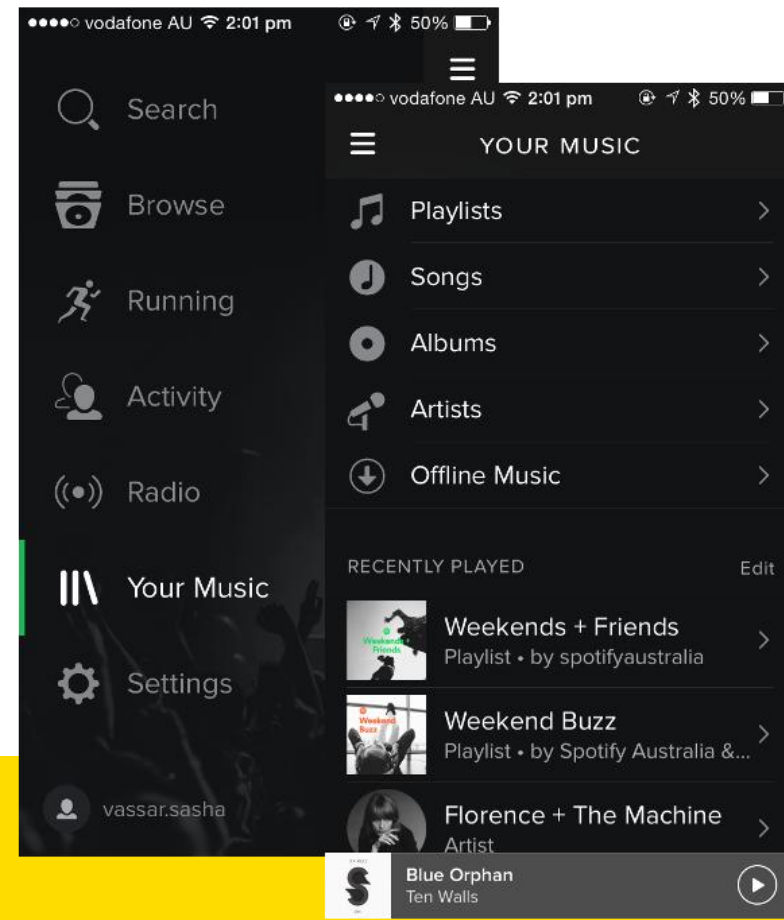
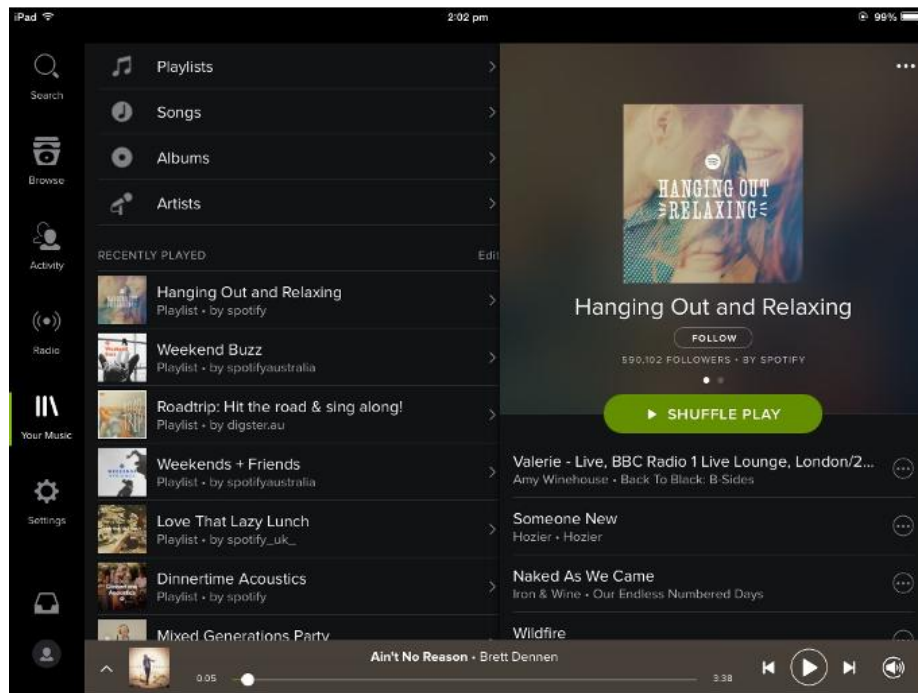
Consistency

- Google G Suit



Consistency

- iPad vs iPhone – same app



Consistency

Internal and External

The screenshot displays the Airbnb homepage with search filters set to 'Anywhere', 'Any week', and '2 guests'. The search results are categorized by 'Lakefront' and show four properties in Australia. Each property listing includes a heart icon, a star rating, the location, dates, and total price.

Property Image	Location	Distance	Dates	Rating	Total Price
	Toukley, Australia	74 kilometres away	25 – 30 July	4.82 ★	\$5,079 AUD total
	Coal Point, Australia	99 kilometres away	7 – 12 July	5.0 ★	\$4,896 AUD total
	Budgewoi, Australia	79 kilometres away	13 – 19 July	4.89 ★	\$908 AUD total
	Toukley, Australia	74 kilometres away	1 – 6 July	4.77 ★	\$4,160 AUD total

Below the main grid, there is a section titled 'Stay in Central Coast and Smiths Lake' with a price of \$2,363.00 AUD total and a 'Show map' button. The footer contains copyright information for 2022 Airbnb, Inc., and links to Privacy, Terms, Sitemap, and Destinations. The language is set to English (AU), the currency is AUD, and there is a link to Support & resources.

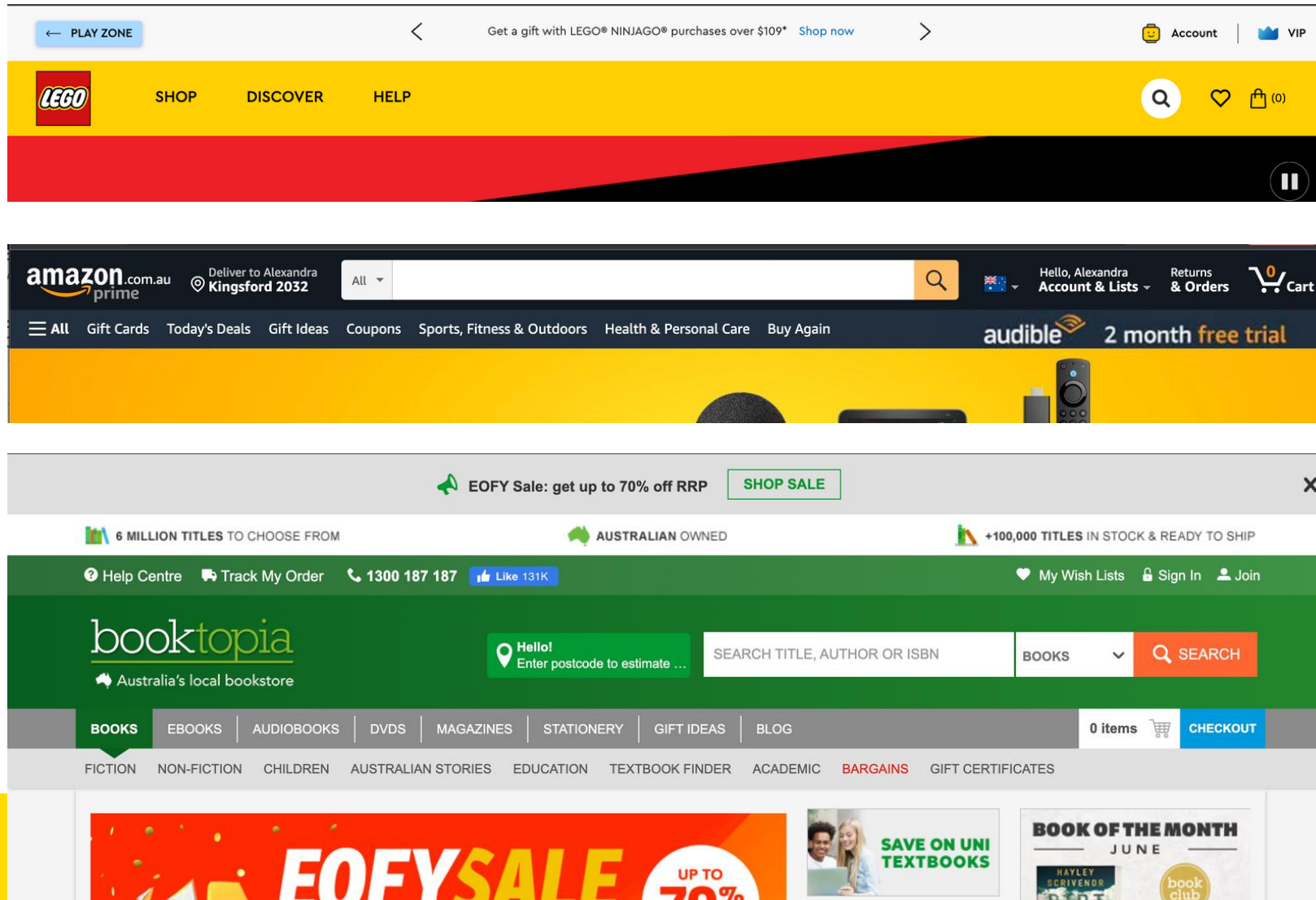
Consistency

External – Jakob's Law

- Users spend most of their time on other sites. This means that users prefer your site to work the same way as all the other sites they already know.

Consistency

External



Consistency

Applied to game design

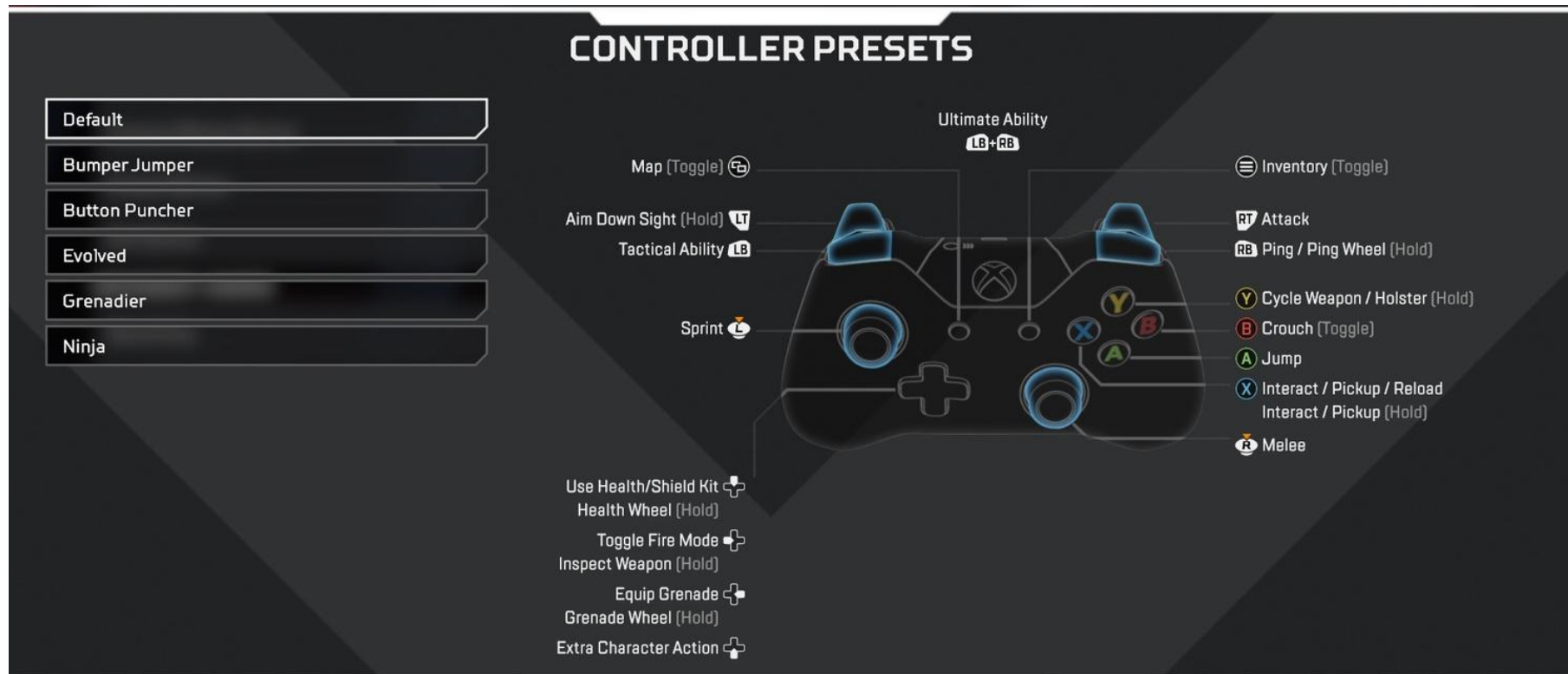


Image credit: <https://www.metabomb.net/off-meta/gameplay-guides/apex-legends-controls-pc-playstation-4xboxone>

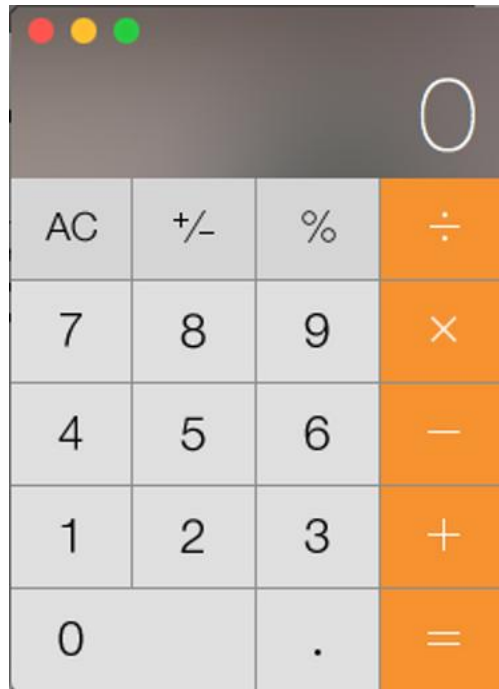
Match Between System and Real World

Real world metaphors

- Familiar concepts to the target audience
- “Speak the user’s language”
- Avoid system-oriented jargon
- Use metaphors wisely
 - understandable, applicable, translatable
- Workflow is reflected in the system

Match Between System and Real World

Skeuomorphic Design



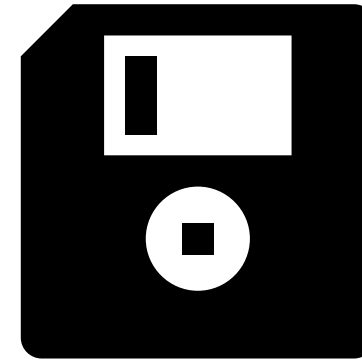
Match Between System and Real World

we keep things for consistency



Bill Gross @Bill_Gross · Oct 17

In the "I'm getting old" department..., a kid saw this and said, "oh, you 3D-printed the 'Save' icon."



Feedback!

- What is the system doing now?
- Keep the user informed
- Show status with minimal delay
- Make the feedback appropriate, meaningful

Feedback!

booktopia

Australia's local bookstore

SHOPPING CART

ADDRESS DETAILS

DELIVERY OPTIONS

PAYMENT AND REVIEW

Shipping Address

Order Summary

zid	name	Question	group	mark	comment
-----	------	----------	-------	------	---------

New Tab

×

+

←

→

↻

Search Google or type a URL

Feedback!



Feedback!

In games ... haptic feedback

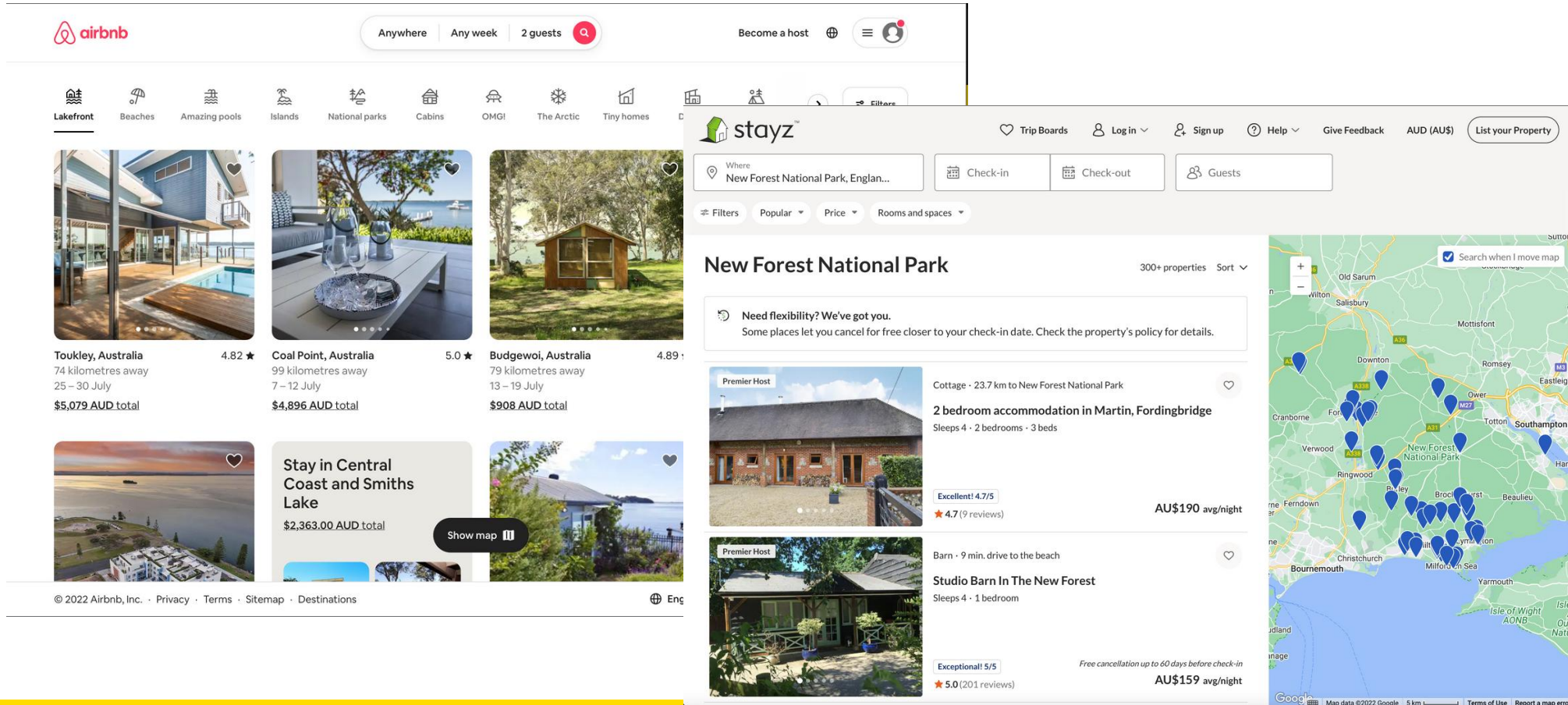


Aesthetic and Minimalist Design

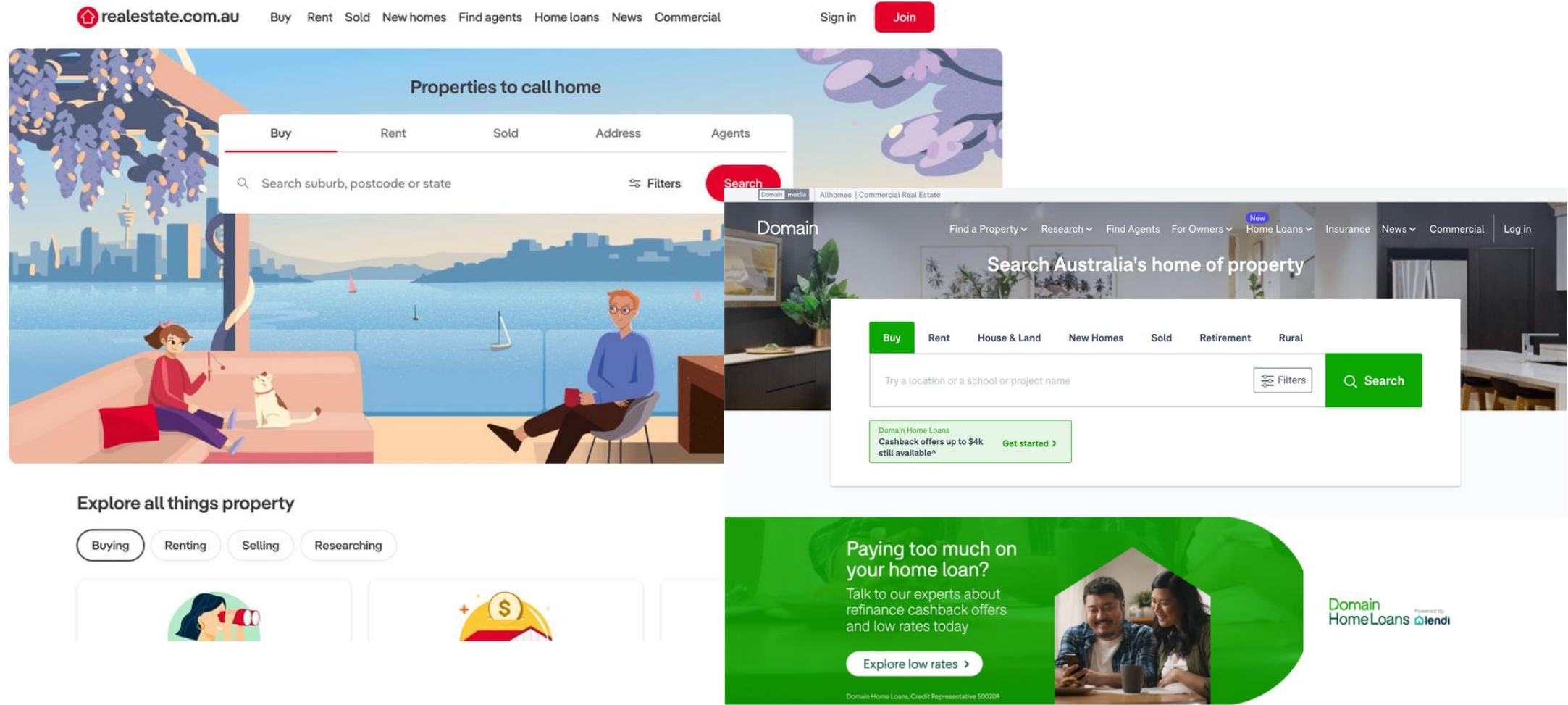
UI

- Avoid providing information that is irrelevant or rarely needed
 - Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.
- How many steps does it take to achieve a task? Do you have any unnecessary steps?

Aesthetic and Minimalist Design



Aesthetic and Minimalist Design



Aesthetic and Minimalist Design

Aesthetic - Usability Effect

- Users often perceive aesthetically pleasing design as design that's more usable.

Aesthetic and Minimalist Design

In games



Aesthetic and Minimalist Design

In games



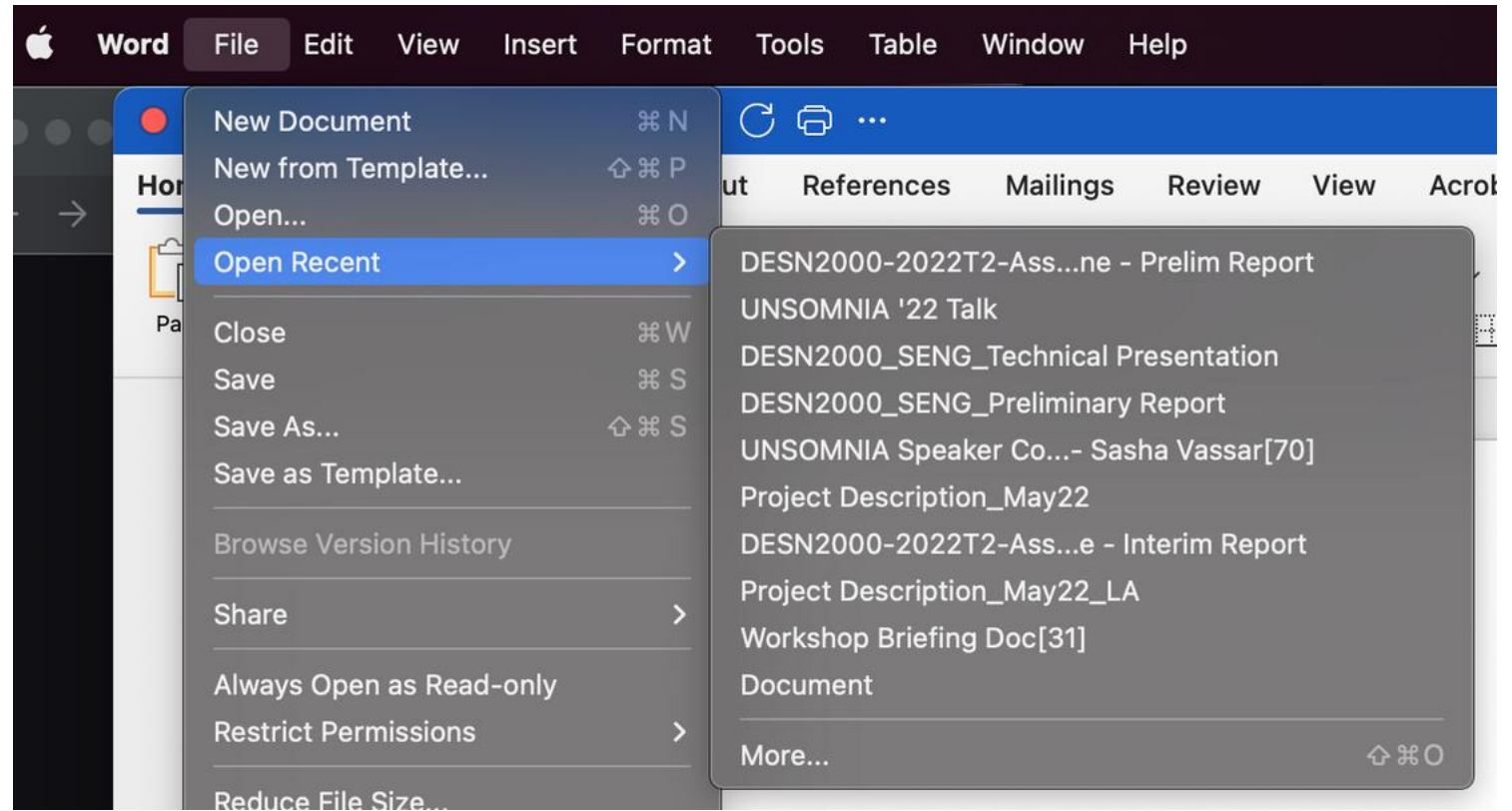
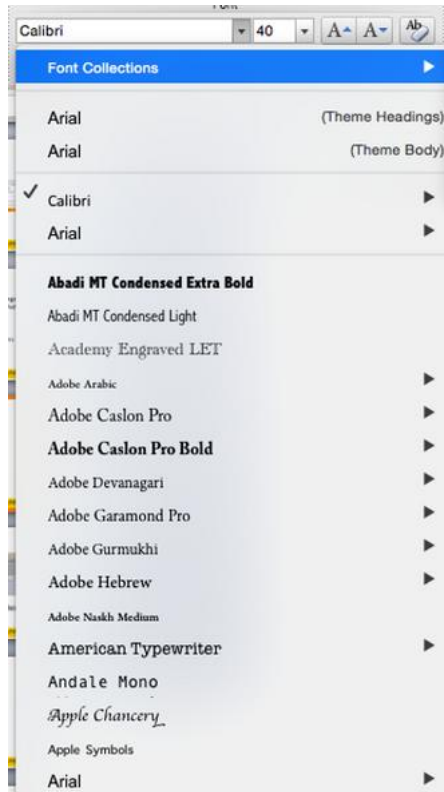
Image credit: <https://gamegetaway.wordpress.com/2016/03/30/world-of-warcraftfound-my-screenshot-of-1-minute-before-the-wow-beta-ended-when-theyreleased-massive-infernals-in-all-the-towns/>

Recognition rather than Recall

UI

- Humans are better at recognising from a range of options than trying to ask them to recall what the options might be
- Make things visible so the user can decide from options presented to them rather than remembering what the (non-displayed) options are

Recognition rather than Recall



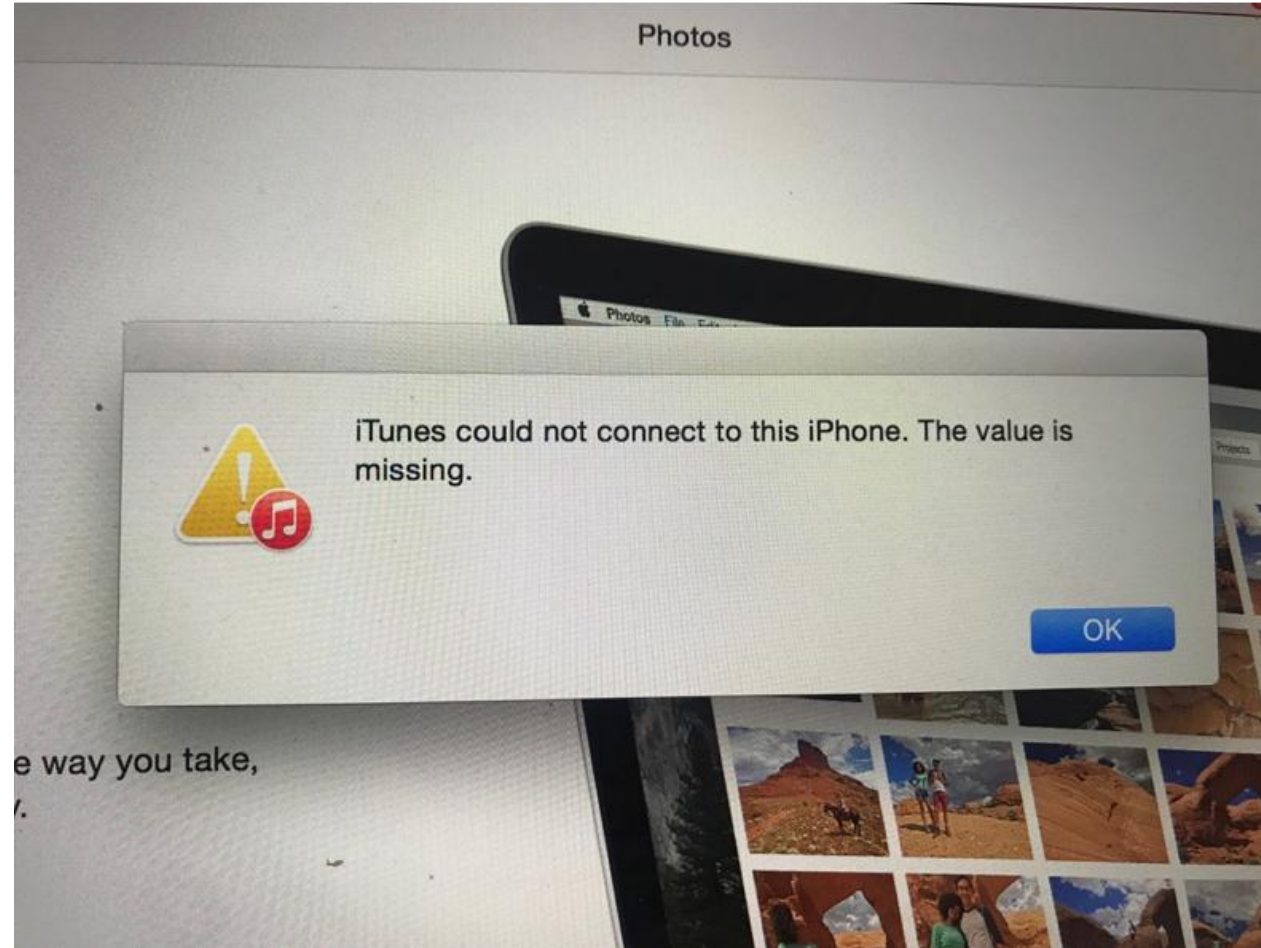
Error Diagnosis and Recovery

UI

- Error messages are presented using plain language that the user can understand
- They can comprehend that they have encountered an error
- The error is described
- There is a suggested solution to recover from the error that is also in plain language

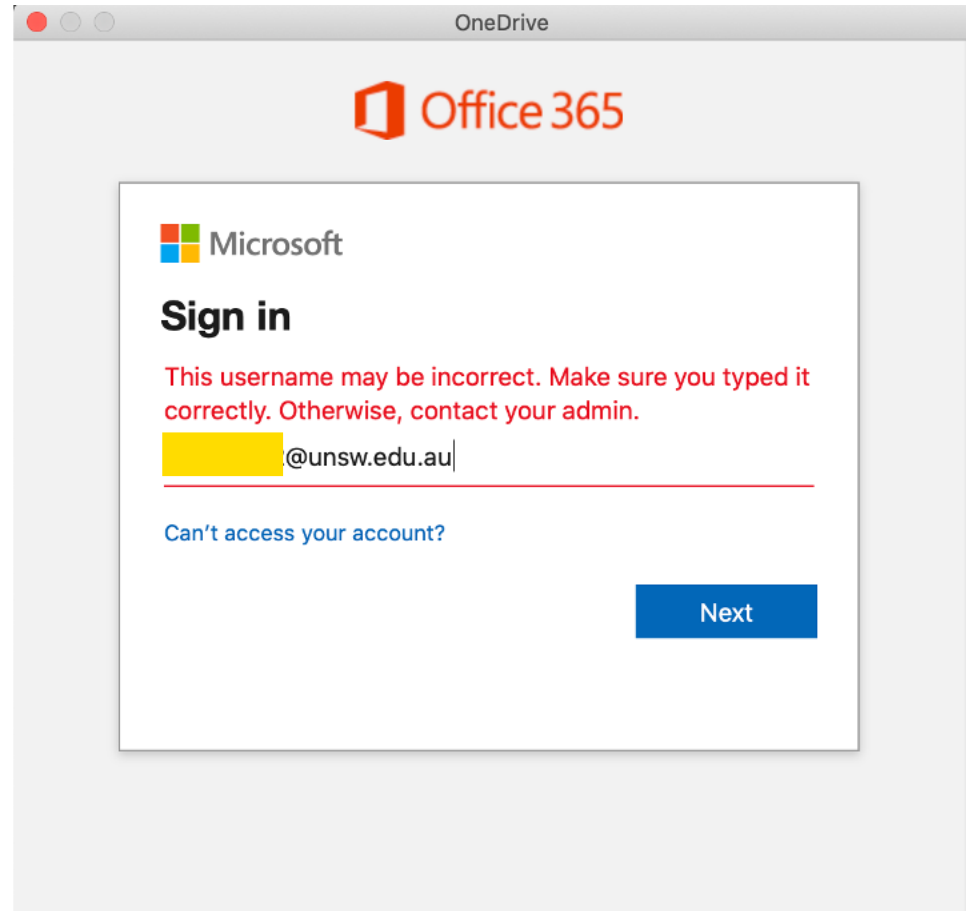
Error Diagnosis and Recovery

UI



Error Diagnosis and Recovery

UI



Error Diagnosis and Recovery

UI

Add members to the TUT_F11A_4284 channel

Students Teachers

You'll need to activate your team to add students to it.

Something went wrong.

Retry all ↺

AM

The user is not a member of the team.

Retry ↺ ×

Error Diagnosis and Recovery

UI

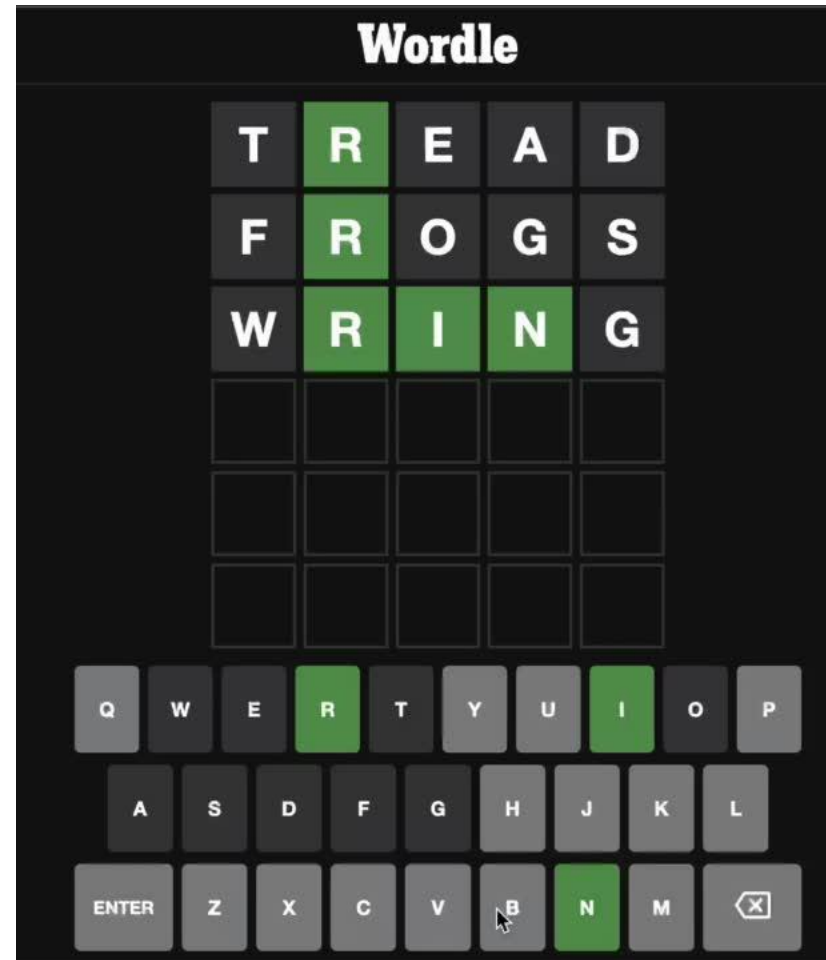
Course	COMP3511 Human Comput	GRD	
Term	Summer Term 2020	Summer Teaching Period	
Faculty	Faculty of Engineering	School	School of Computer Sci & Eng
Campus	KENS - Sydney	Class	All students
Enrolments	71	Item	All selected (6) Confirm

An unknown error has occurred.

Close

Error Diagnosis and Recovery

In games



Not in word list

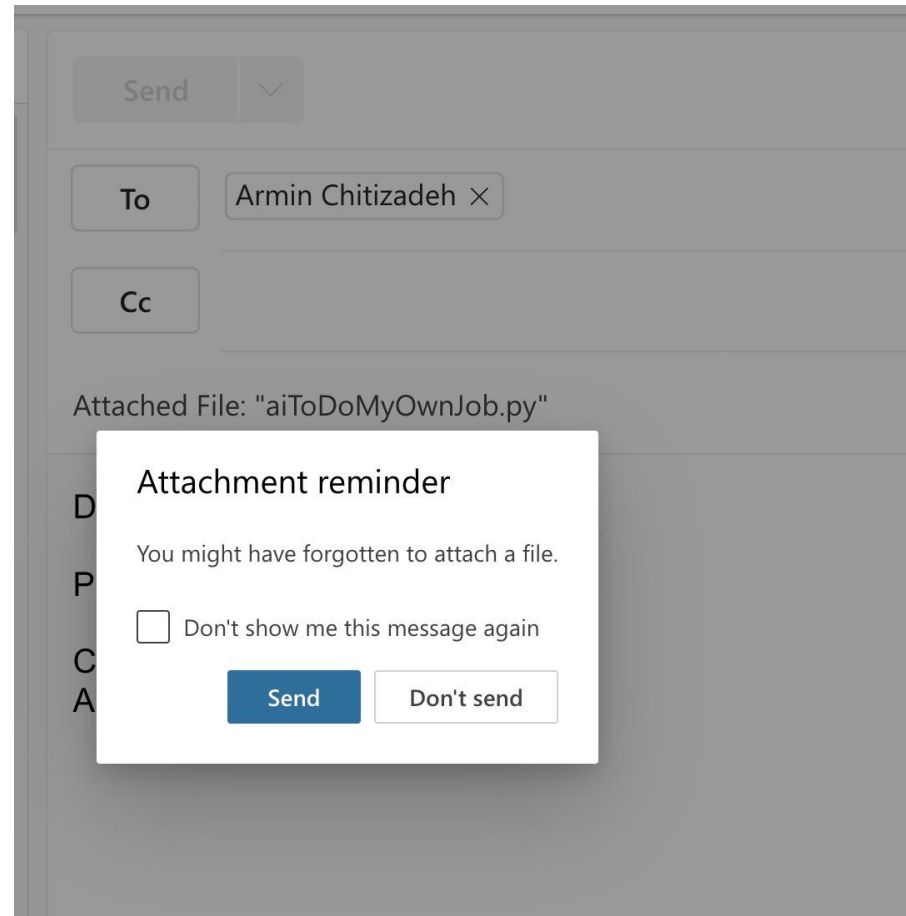
Error Prevention

UI

- Stop errors from occurring in the first place
- Proximity of controls to each other
- Providing clear labelling or meaningful terminology
- Examples along side where input needs to be made

Error Prevention

UI



Error Prevention

UI

Stays Experiences Online Experiences

Where
Search destinations

Date
Add when you want to ...

Who
2 guests

Search

< June 2022 >

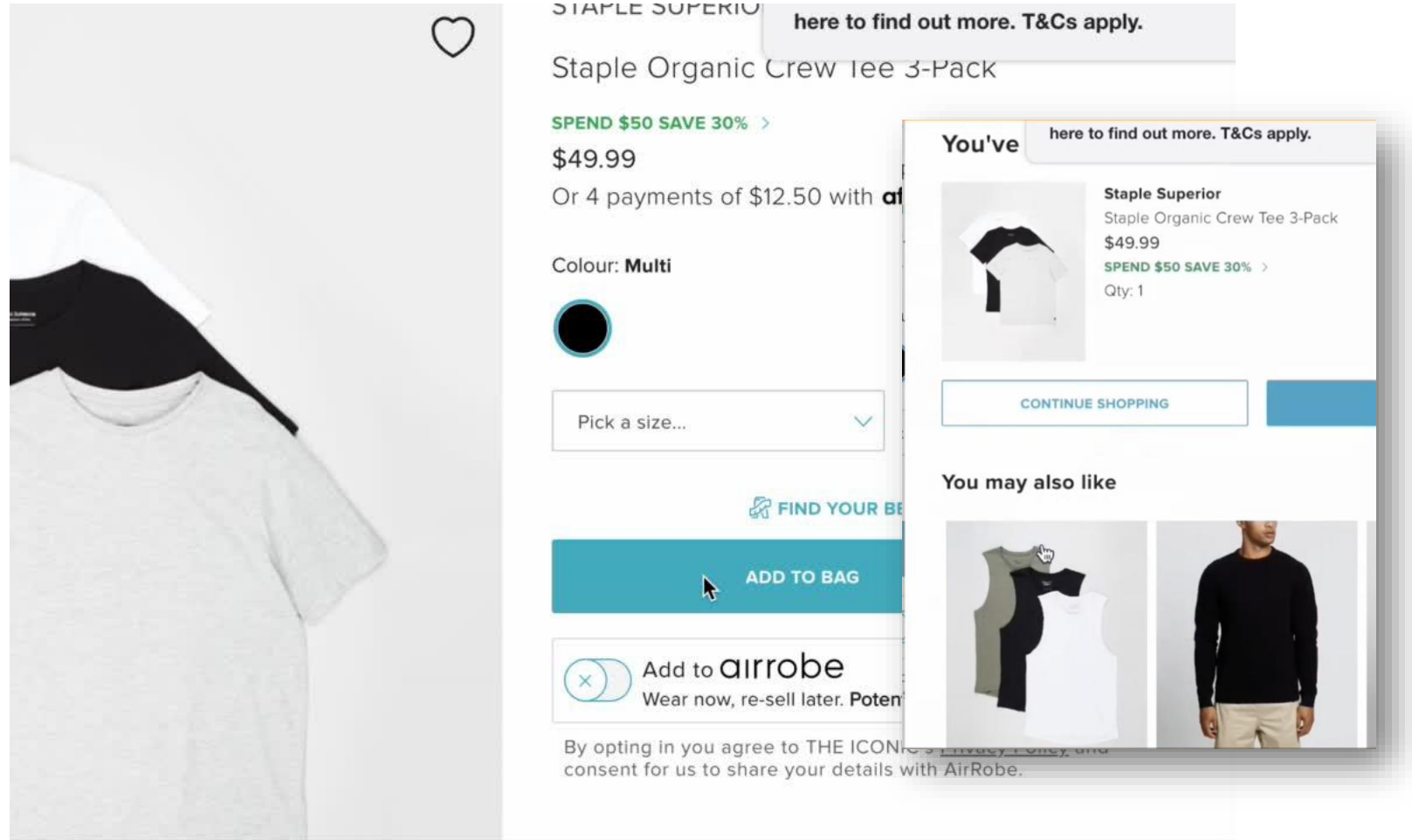
Su	Mo	Tu	We	Th	Fr	Sa
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

< July 2022 >

Su	Mo	Tu	We	Th	Fr	Sa
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3	4	5	6	7	8	9
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17	18	19	20	21	22	23
24	25	26	27	28	29	30
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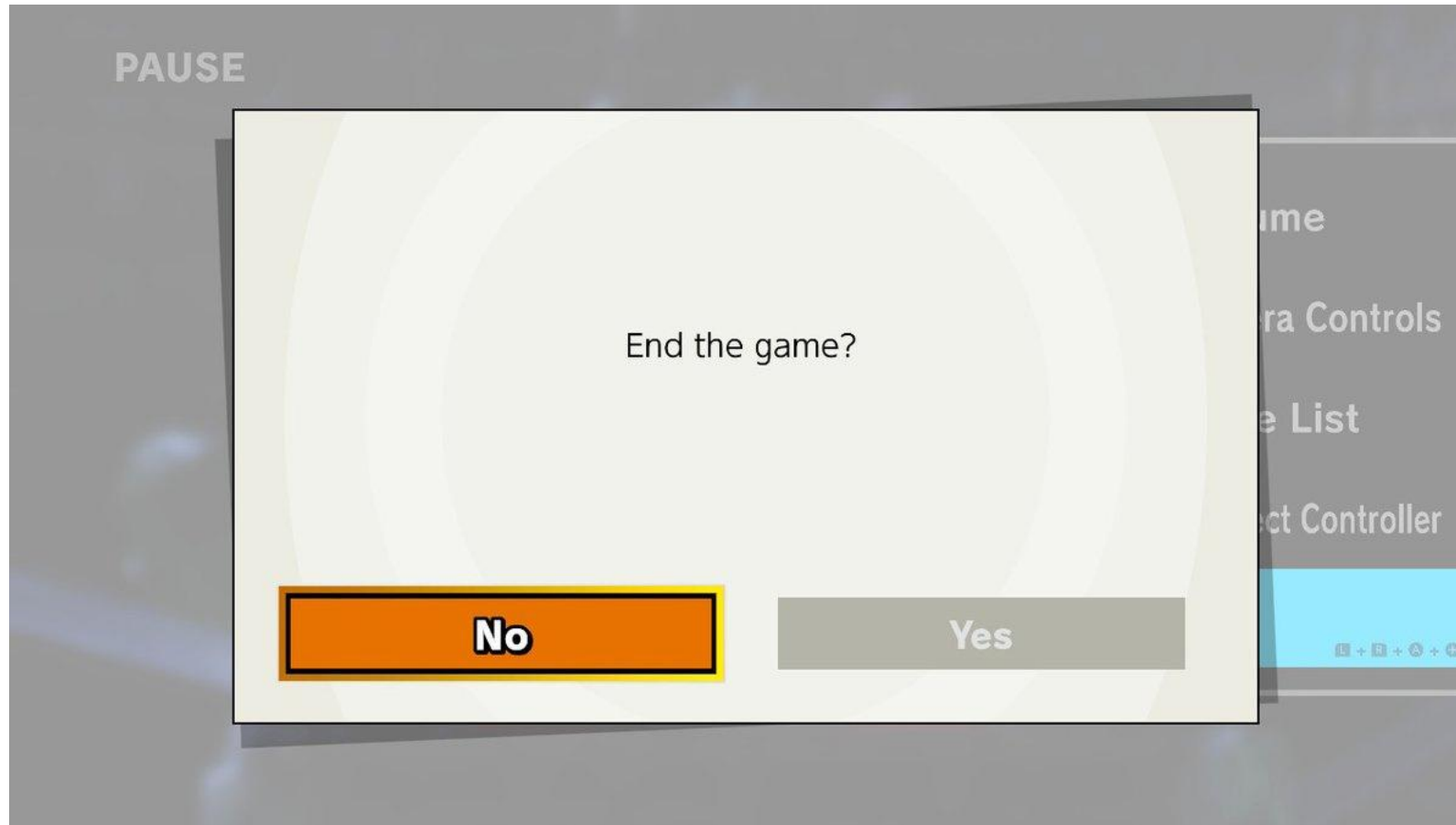
Error Prevention

I



Error Prevention

In games ...



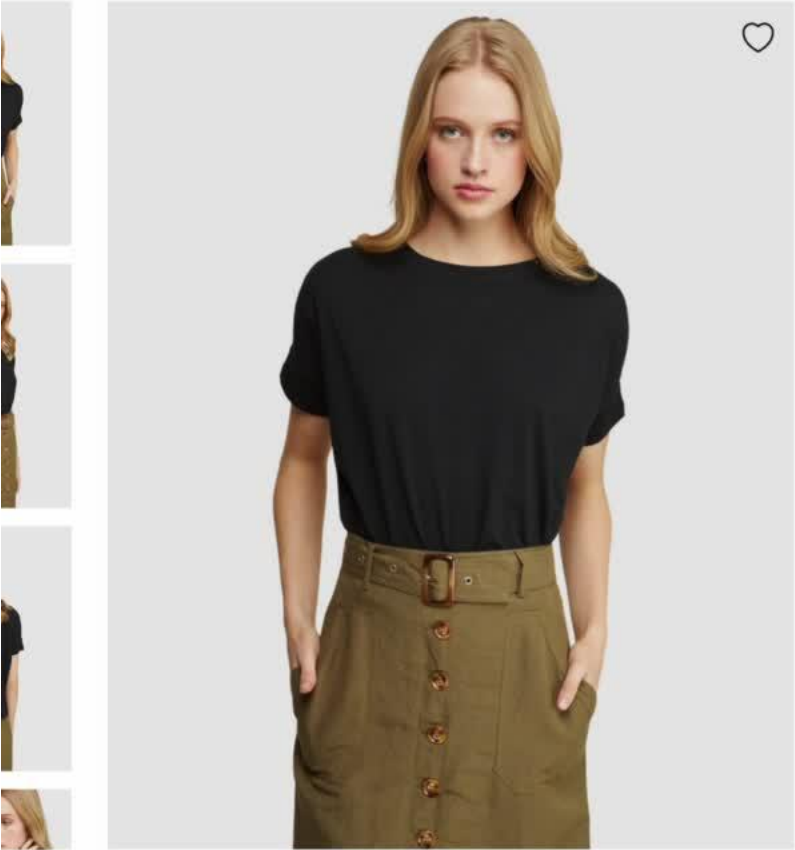
Help and Documentation

- UI
 - Better if system can be used without help
 - Help can't be too bulky
 - Information that can be easily searched
 - Provides help in a series of concrete steps that are easily followed

Help and Documentation

UI

5 / T-SHIRTS & SINGLET



OXFORD > [+ Follow](#)

Aria T-Shirt

40% OFF AT CHECKOUT >

~~\$69.00~~

\$59.00

Or 4 payments of \$14.75 with **afterpay**

Colour: **Black**



Pick a size...

[FIND YOUR BEST FIT](#)

ADD TO BAG

 **Add to airrobe**
Wear now, re-sell later. Potential value

By opting in you agree to THE ICONIC's [Privacy Policy](#) and consent for us to share your details with AirRobe.

Size & fit
Model wears: Size 8

Care
Cold hand wash, line dry, warm iron. Do not tumble dry. Iron inside out to retain the shape. Do not bleach.

Delivery
This item is sent directly from our partner and will arrive **separately** if ordered with other items.

Clothing Shoes

SIZE GUIDE - OXFORD

Free Returns
If it doesn't quite fit, send it back using our free return shipping!

Clothing

Size	AU Size	Bust (cm)	Waist (cm)	Hip (cm)
6	XS	76-78	60-62	84-86
8	SM	79-83	63-67	87-91
10	SM	84-88	68-72	92-96
12	ML	89-93	73-77	97-101
14	ML	94-96	78-80	102-104

Shoes

Size	AU Size	UK Size	USA Size
36	5	3	5
37	6	4	6
38	7	5	7
39	8	6	8
40	9	7	9

 **Help**

Help and Documentation

games

STARTER GUIDE



DRIVER'S MANUAL

Braking



Braking is under-appreciated! If it seems like you're about to go out of control and hit a wall or

an obstacle on the track, by tapping the brakes you can regain some maneuverability to get away from danger—at the cost of speed.

An advanced technique called Drift Braking, which is useful mainly during 200cc races, involves applying the brake during sparking drifts—enough brake to slow the lateral movement of the drift and keep the car on-track, but not so much brake that the drift dissolves into a mere slow turn. This technique is paramount to 200cc race success, can be helpful on the nastiest corners in 150cc races, and can be disregarded at the 100cc level or lower.

Jump



The drift button doubles as a jump button. Jumping won't take you over items on the

road, but it can lift you over small gaps in the road. You can cut corners that overlook bottomless pits this way, but it can be quite risky to try.

Drift

Drifting allows you to tackle turns without losing speed. To enter a drift, point the nose of your vehicle into a turn, then hold R. The vehicle will hop into the turn, lose traction, and continue sliding with the same inertia as before, though the nose is pointed inward. The sharper your vehicle's nose is turned in while drifting, the faster you'll build up blue sparks, then orange, and eventually purple, allowing you to set off a Mini-Turbo, or a Super/Ultra Mini-Turbo. When R is released to activate a built-up turbo boost, traction is suddenly restored and the vehicle will cease lateral travel and surge forward toward where the nose is pointed. Note that the various vehicle chassis heavily affect how you drift.



Sparks and Mini-Turbo



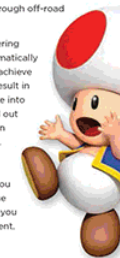
While drifting would be useful anyway for the way it lets you tackle turns, it's even more useful for all

the extra speed it can create. As you sustain a drift, sliding sideways through a turn while pointing the nose inward, sparks will begin to build up under the tires. Releasing a drift once sparks are happening results in a Mini-Turbo boost! It should soon become second-nature to be drifting into Mini-Turbo boosts off of most turns, only resorting to standard turn-and-brake cornering if the turn is too sharp or short for a solid drift, or the kart isn't going fast enough to initiate a drift (like after eating an enemy's item or smacking a wall).



The color of the sparks indicates how long the drift has been held, and how powerful the follow-up boost will be. Drift until blue sparks appear for a Mini-Turbo boost. Drift even longer for orange sparks, leading to a Super Mini-Turbo, which lasts about twice as long as a Mini-Turbo. On the longest turns, you can even drift long enough to build up to purple sparks, resulting in an Ultra Mini-Turbo boost, basically a free Mushroom! (However, unlike a Mushroom, it's important to note that Mini-Turbos won't carry you through off-road terrain like Mushrooms will.)

Purple sparks can only occur if Steering Assistance is disabled. It's not automatically better to hold a drift longer just to achieve orange or purple sparks—this can result in a risky or inefficient line, as you slide into the far walls of a turn trying to hold out for the next level of Mini-Turbo when basic blue sparks would've sufficed. Holding a drift longer for better turbo can make your movement more predictable in turns, leaving you open to item attacks or drafting. Use whichever level of Mini-Turbo gives you the best racing line in a given moment.



The Drivethrough



Mid-air from the glide ramp, you have some work to prepare for the end of the lap. On Lap 1, you can snag your final Coins to reach ten while in mid-air.



As you're gliding down, avoid the ramp at the center of the course. As you pass up the ramp, drift toward the sand and the pipes on the inside of the final turn. You should be able to pull off a Mini-Turbo just as you reach the sand.



The scattered pipes on the sand can be devastating to hit during a Time Trial, but the best time to use Mushrooms is to blitz through this pipe forest across the sand. By doing this, you cut the corner and practically have a straight shot at the finish line!



Time Trial Mode



Nintendo's Best 200cc Time Trial Time: 1:21.984
Player: Nin'Sophia

Racer: Gold Mario
Body: Gold Standard
Wheels: Gold Tires
Glider: Gold Glider

Nintendo's Best 150cc Time Trial Time: 1:53.191

Player: Nin'Chris
Racer: Mario
Body: Sports Coupe
Wheels: Standard
Glider: Super Glider

Image credit: <https://my.nintendo.com/rewards/9fd236f5dcda94c4>

Some UX Considerations

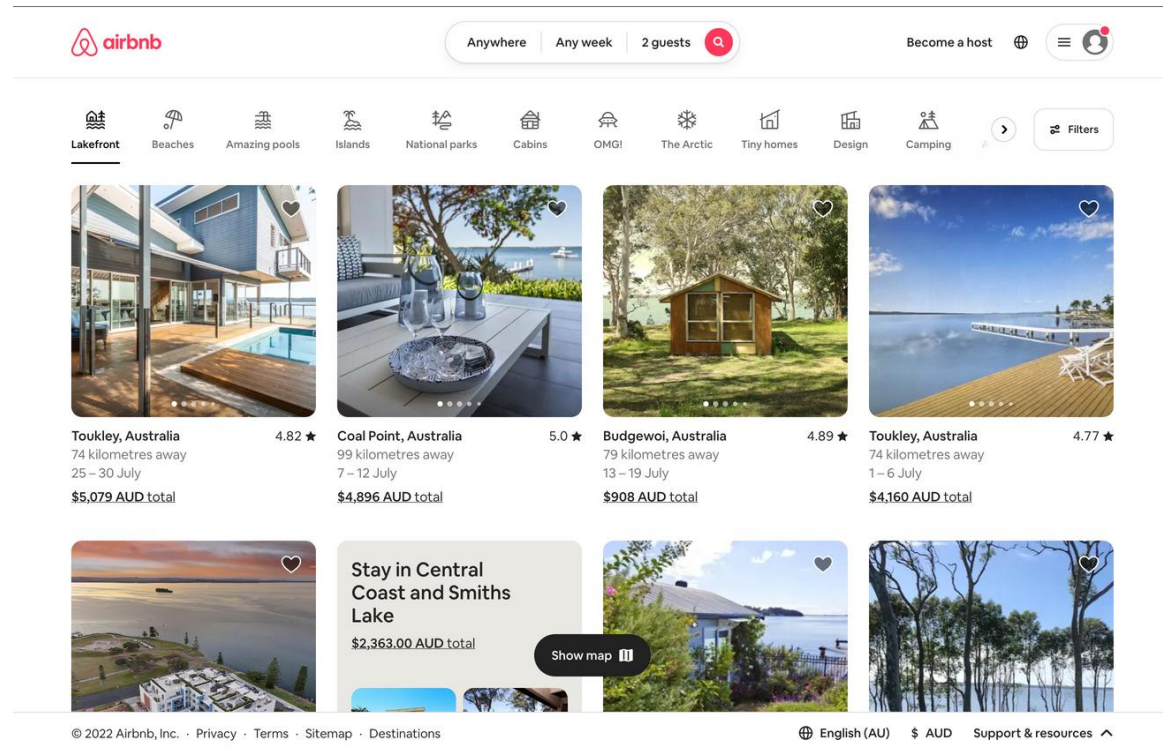
Fitt's Law - Human Predictive Model

- The time to acquire a target is a function of the distance to and size of the target
 - The time it takes someone to select an object depends on how far they are from the object and the size of that object



Some UX Considerations

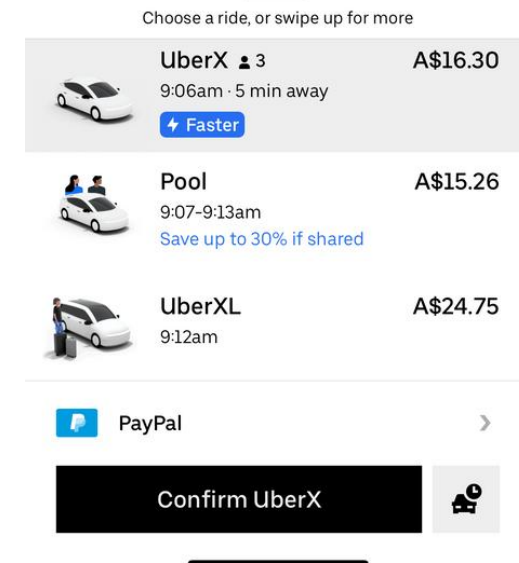
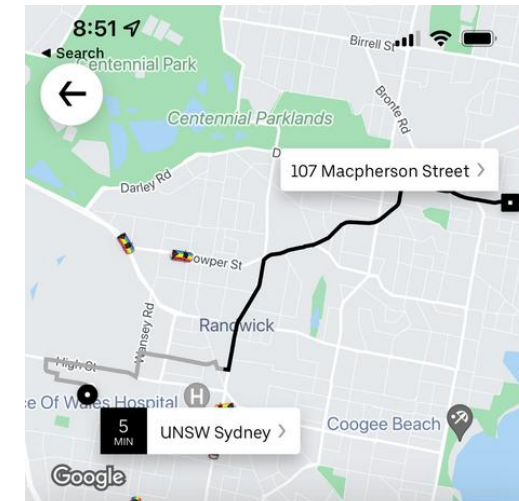
Fitt's Law - Human Predictive Model



Some UX Considerations

Goal-Gradient Effect

- Closer you are to completing a task, the faster you work towards reaching it
 - Think progress bars!



Some UX Considerations

Hick's Law

- The time it takes to make a decision increases with the number and complexity of choices.
 - Break up complex tasks
 - Progressive onboarding
 - Highlight recommended options

Some UX Considerations

Hick's Law

The screenshot shows the Amazon Australia product page for Mario Party Superstars. The page layout includes a top navigation bar with the Amazon logo, delivery location (Kingsford 2032), search bar, and account links. Below this is a secondary navigation bar with categories like Video Games, Deals, and Best Sellers. The main content area displays the product image, title, brand, platform, and price. A yellow rounded rectangle highlights the purchase options section, which includes a quantity selector, 'Add to Cart' and 'Buy now' buttons, and a 'Secure transaction' link.

amazon.com.au prime Deliver to Alexandra Kingsford 2032 Video Games

Video Games Deals Best Sellers Pre-orders & New Releases PlayStation Xbox One Nintendo Switch PC Games Accessories

Video Games > Nintendo Switch > Games

Mario Party Superstars - Nintendo Switch
Brand: Nintendo
Platform: Nintendo Switch
★★★★★ 88 ratings
Amazon's Choice for "mario party superstars"

-20% \$63⁹⁹
RRP: \$79.95
prime Same-Day

About this item

- 100 classic minigames return from the Nintendo 64 and Nintendo GameCube games and more
- Try your hand at a Superstar collection of minigames from throughout the Mario Party series; Whether you are saving up coins for stars in board game mode or practicing in free play, there is nothing quite like being the last

\$63⁹⁹
prime Same-Day
FREE delivery available Today by 9 PM. Order within 2 hrs 8 mins
Deliver to Alexandra - Kingsford 2032
In stock

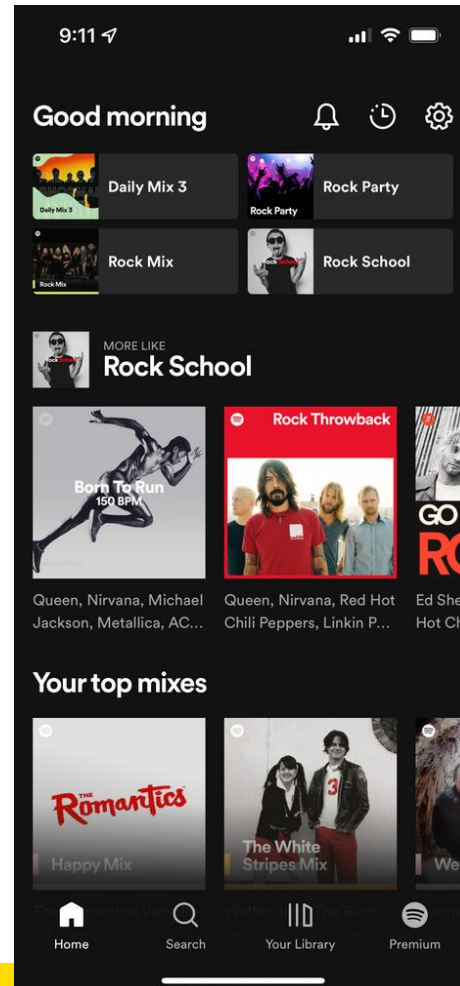
Quantity: 1

Add to Cart
Buy now

Secure transaction
Ships from Amazon
Sold by ACSTechnology

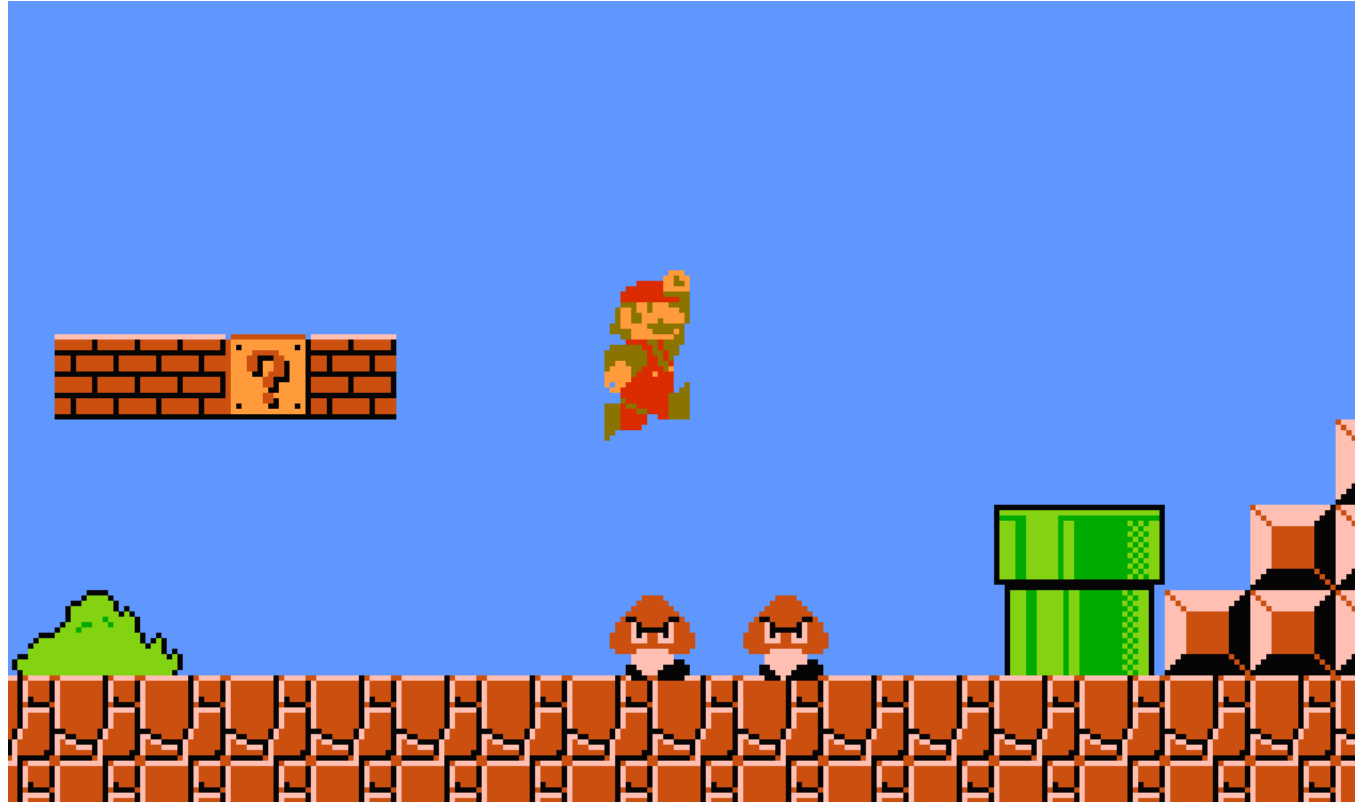
Some UX Considerations

Hick's Law



Some UX Considerations

Hick's Law



Some UX Considerations

Other effects

- Will be covered in Week 7, as part of visual hierarchy and visual design lecture

Design Patterns

UI

- Reusable/recurring components that are used to solve common problems in user interface design.
- Basically, blueprints that allow designers to choose the best and most commonly used pattern for the specific context
- Each pattern typically contains:
 - The problem a user is facing
 - The context of the problem
 - The principle/guideline involved
 - The proven solution

Design Patterns

UI

- Only helpful when based on specific design cases
- Core functions:
 - Input/Output Navigation
 - Content Structuring
 - Social Sharing
 - Onboarding

Design Patterns

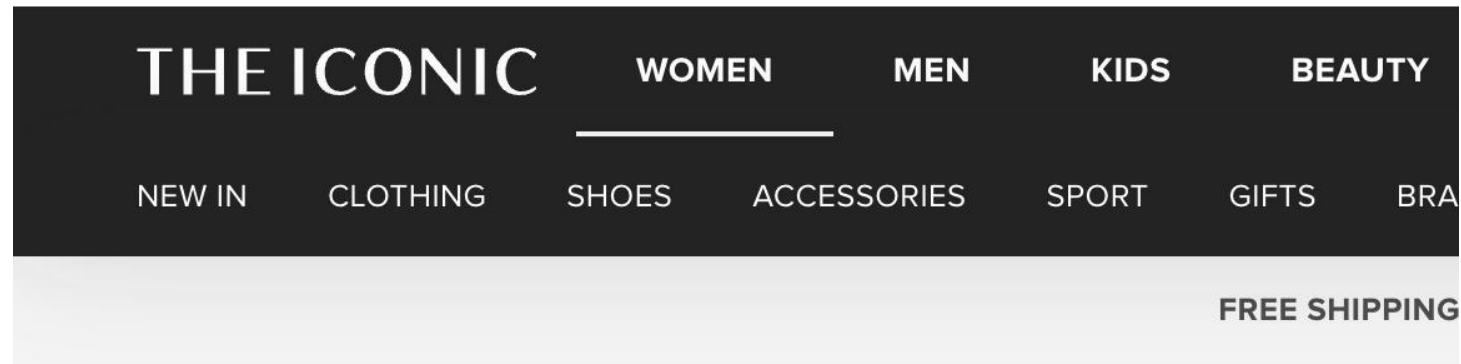
Some Common Patterns

- Breadcrumbs
- Lazy Registration
- Reaction
- Calendar Picker
- Steps Left

There are so many: <https://ui-patterns.com/patterns>

Design Patterns

Breadcrumb Navigation Pattern



WOMEN / CLOTHING / TOPS



Design Patterns

Lazy Registration Onboarding Pattern

[← Continue Shopping](#)

THE ICONIC

 BAG (2 ITEMS)

 DELIVERY





Low stock

Nude Lucy
Nude Classic Knit
~~\$69.95~~ **\$41.00**
SALE

[Move to wishlist](#)
[Remove](#)

Qty

1

Size

M

Hi There!

Please login to continue to delivery and payment.

[LOGIN](#)

New to THE ICONIC? [CREATE ACCOUNT](#)



Low stock

Staple Superior
Staple Organic Crew Tee 3-Pack
\$49.99
SPEND \$50 SAVE 30% >

[Move to wishlist](#)
[Remove](#)

Qty

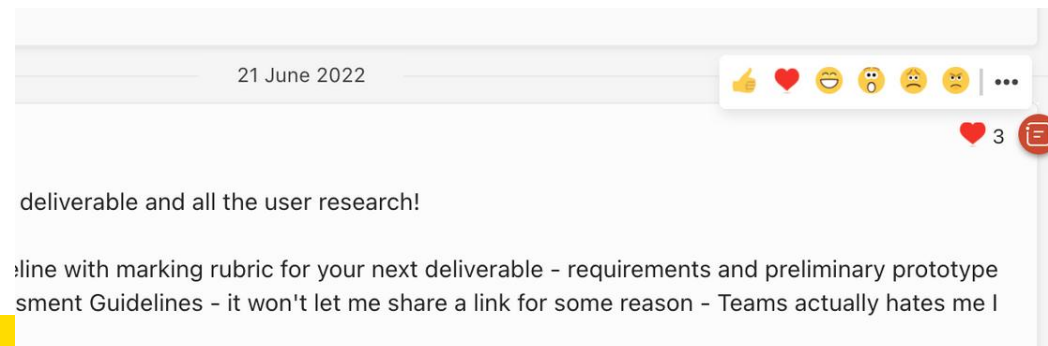
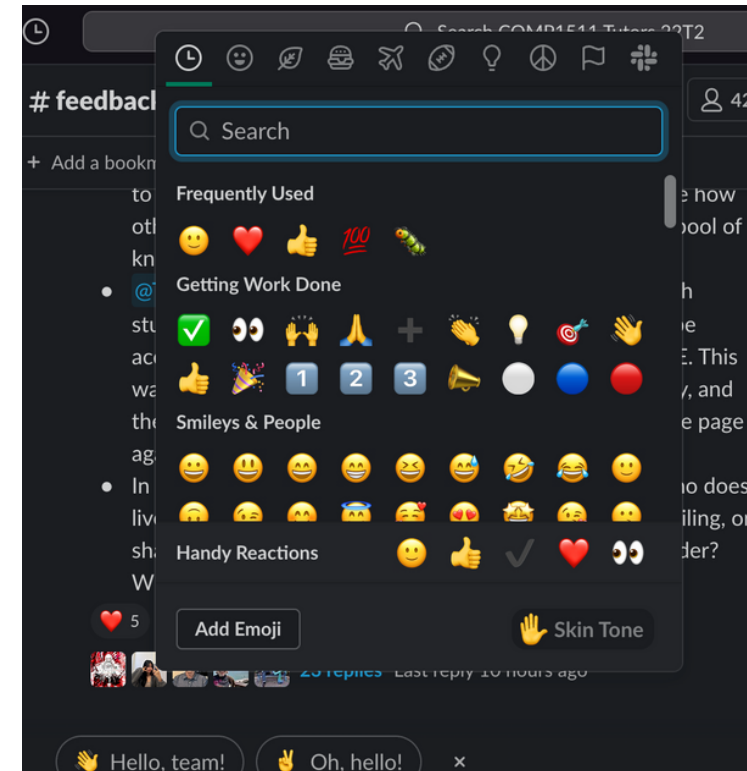
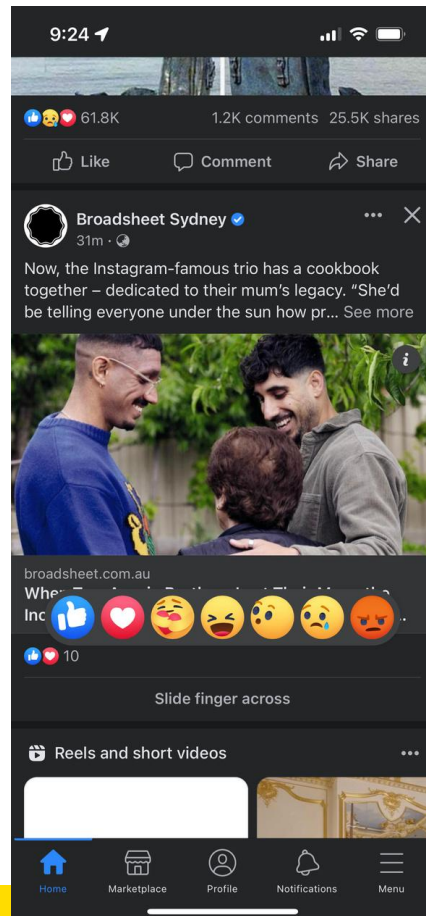
1

Size

XS

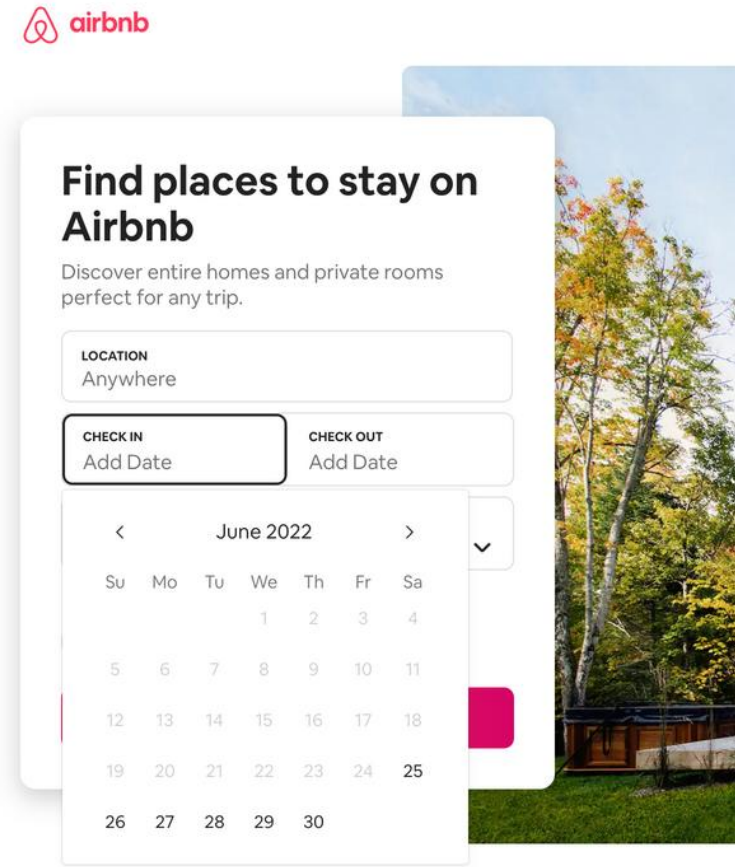
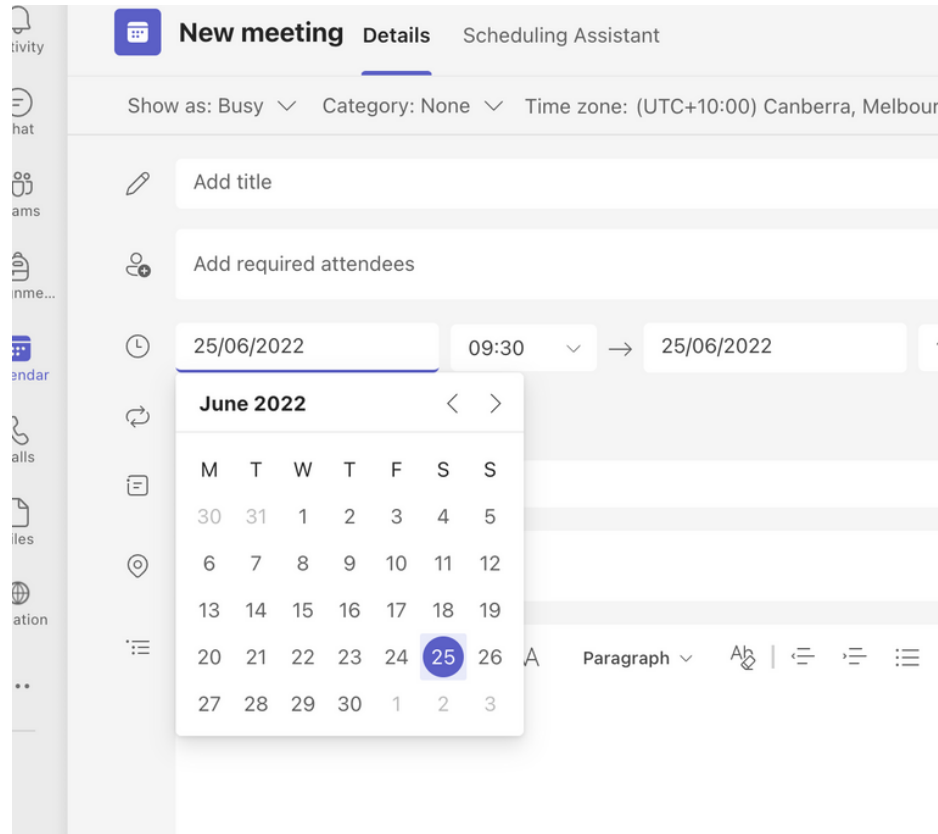
Design Patterns

Reaction Social Pattern



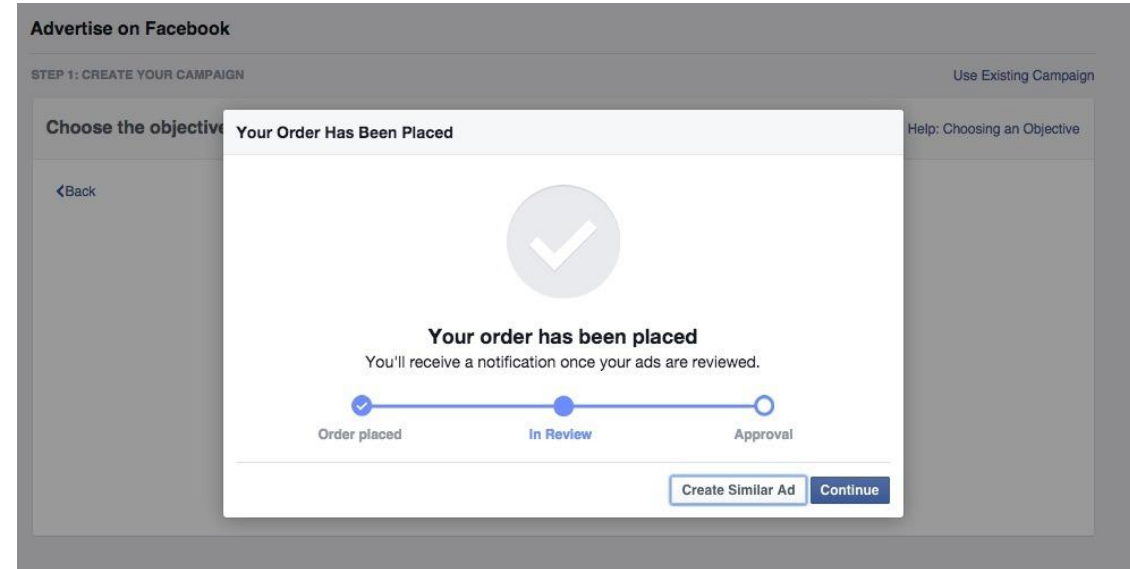
Design Patterns

Calendar Picker



Design Patterns

Steps Left



<https://ui-patterns.com/patterns/StepsLeft>



Design Patterns

Dark Patterns

- Interactions designed to mislead or trick users to make them do something they don't want to do
- Some common patterns:
 - Bait and Switch
 - Sneak into Basket
 - Trick Questions
 - Disguised Ads
 - Forced Continuity
 - Friend Spam
 - Hidden Costs
 - Misdirection
 - Price Comparison
 - Prevention
 - Privacy Zuckering
 - Roach Motel

Design Patterns

Dark Patterns – Bait and Switch



<https://www.cnet.com/tech/services-and-software/microsoft-windows-10-upgrades-getmore-sneaky-pushy/>

Design Patterns

Dark Patterns – Forced continuity

- Sign up for a service with free trial but give your credit card details and then get charged ...

Design Patterns

Dark Patterns – Friend Spam

- LinkedIn case in 2015



BUSINESS • CLASS ACTION

LinkedIn to Pay \$13 Million in Spam Settlement

BY DENVER NICKS 

OCTOBER 6, 2015 7:47 AM EDT

The professional networking site LinkedIn has agreed to a \$13 million out of court settlement in a class action lawsuit that claims reputations may have been **damaged** by multiple emails the company sent on behalf of users.

The suit originated in California in 2013, when LinkedIn users sued the company claiming its “Add Connections” feature hurt their professional reputation by relentlessly messaging their email contacts with requests to connect on LinkedIn, *Fortune* reports. In the complaint, users described being embarrassed by the emails and complained that it was very difficult to stop LinkedIn from sending more emails once the barrage had begun. The settlement affects users who signed up for LinkedIn’s “Add Connections” feature between September 2011 and October 2014.

<https://time.com/4062519/linkedin-spam-settlement/>

Design Patterns

Dark Patterns – Hidden Costs

Mary Poppins
PM, Sydney Lyric, Pyrmont, New South Wales

01:48

ucher

HER ONLY.

prepared cheese & antipasto or dessert
I, you will be provided with a Royal Box &
ox Collection Point at the main Stalls bar

– 0 +

– 0 +

SEC
DRESS

ROW
G

SEAT
40



Order Details ^

Cancel Order

1 Standard Ticket 
AU \$119.90 x 1
Price Category Three, 2. Dress Circle Seating

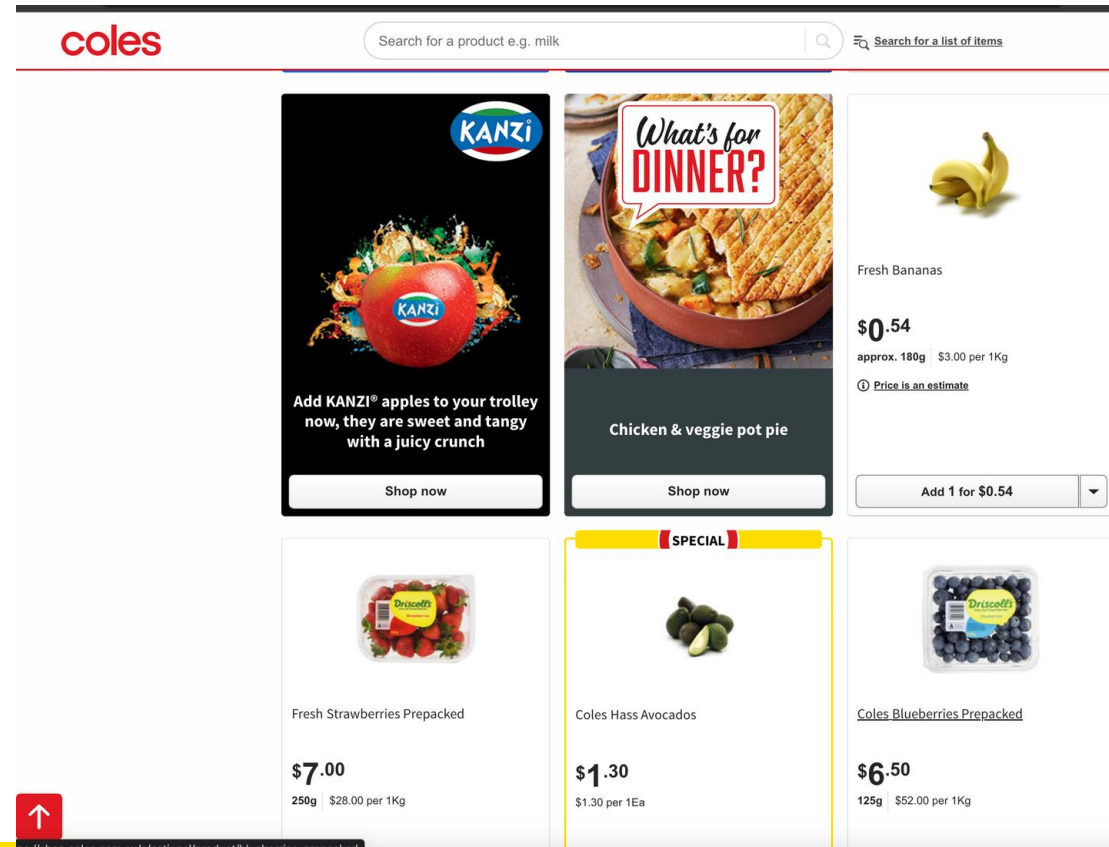
Handling Charge
AU \$9.40

Subtotal
AU \$129.30

Next

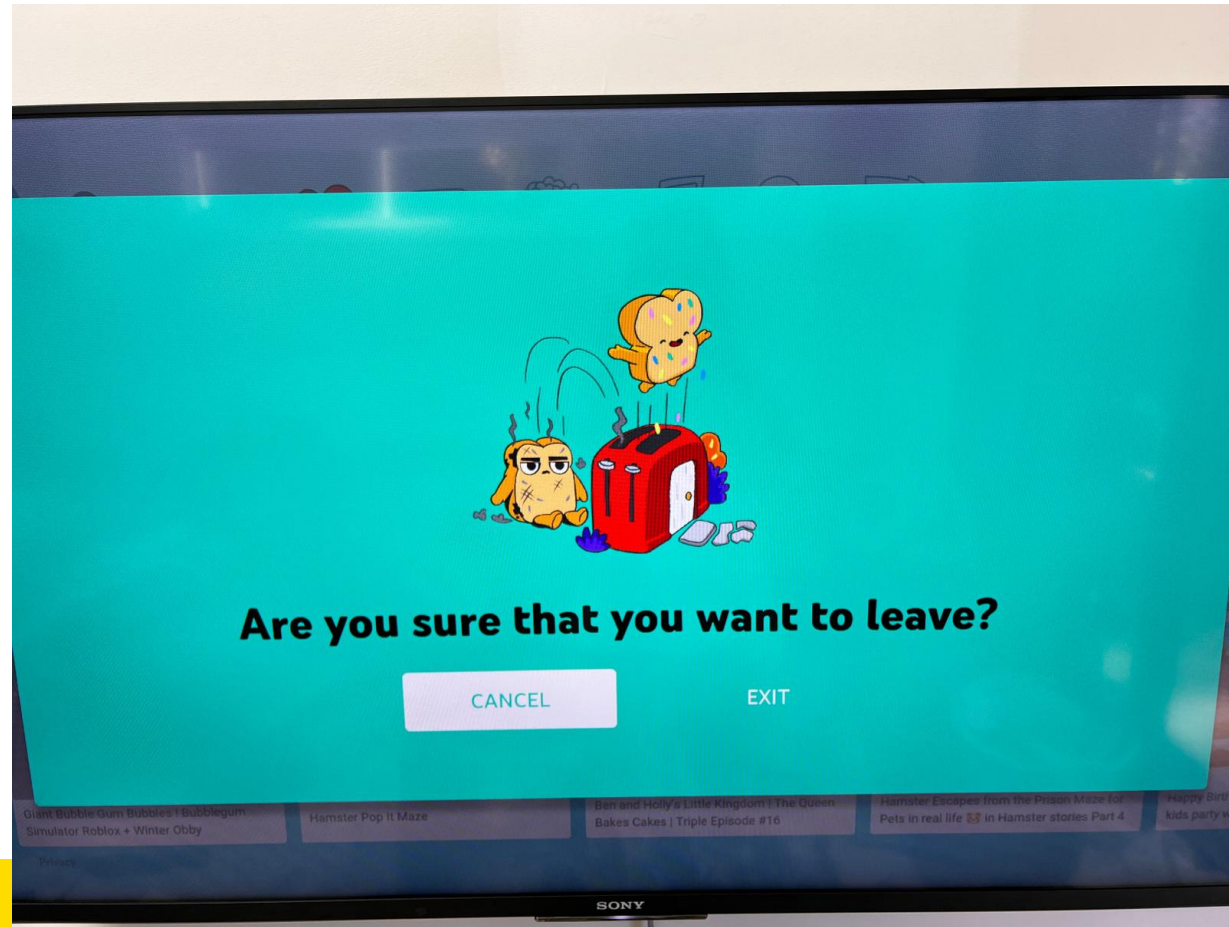
Design Patterns

Dark Patterns – Price Comparison Prevention



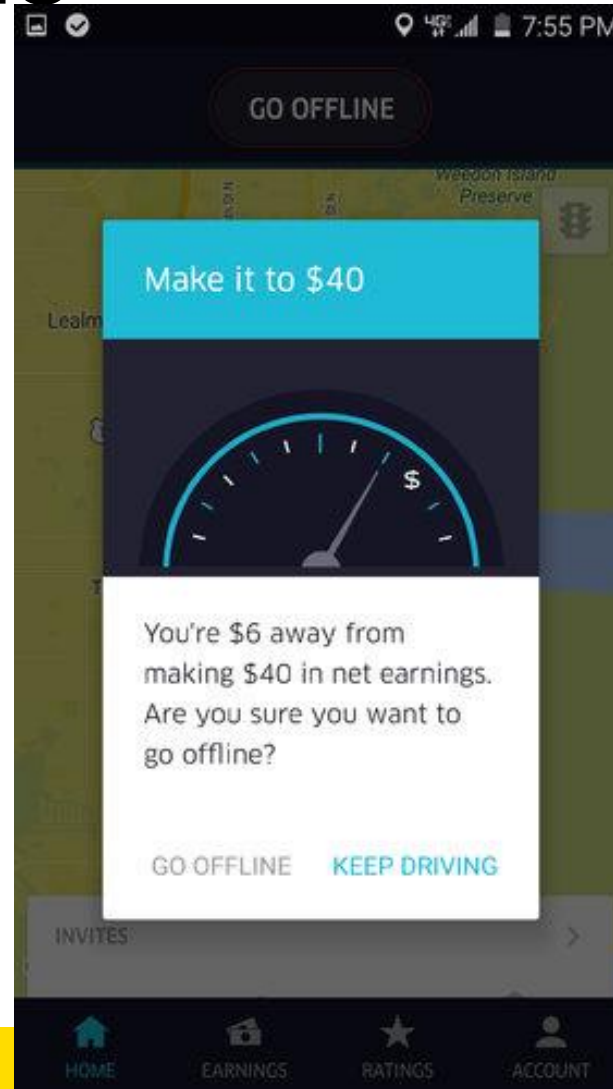
Design Patterns

Dark Patterns – Trick Questions – Kids YouTube



Design Patterns

In-vehicle interface



<https://www.nytimes.com/interactive/2017/04/02/technology/uber-drivers-psychological-tricks.html>

Design Patterns

Dark Patterns in Games

- Different classification in the context of video games
- In this context, a dark pattern is defined as motivational design pattern "used intentionally by a game creator to cause negative experiences for players which are against their best interests and likely to happen without their consent.... (violating) user expectations by encouraging them to give up or jeopardize some resource to an extent that they were not expecting (time, money, social capital)."
- Temporal dark patterns
 - Take more or less time than players expected.
 - The player is being "cheated" out of their time
- Monetary dark patterns
 - Players are being deceived into spending more money than they expected or anticipated or spending money at unexpected moments
- Social capital dark patterns
 - Player's social capital, loosely defined as the value of their social standing and relations, is being risked

Design Patterns

Grind temporal dark pattern



Image credit: <https://www.ign.com/wikis/halo-4/Customization>

Design Patterns

Currency Confusion monetary dark pattern

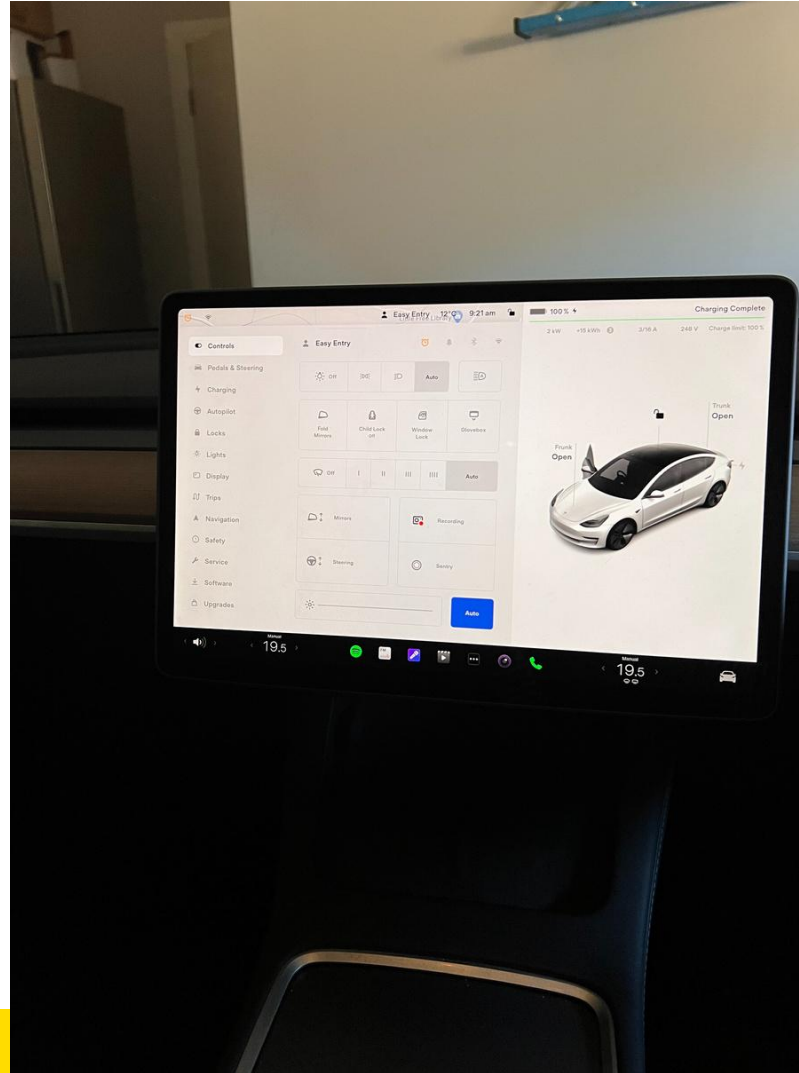


Image credit: https://xbox.fandom.com/wiki/Microsoft_Points

Design Patterns

In-vehicle Interface

What do you think of this interface?



Design Patterns

In-vehicle interface



Next Time

Visual Hierarchy, typography and space

- Visual hierarchy
- Visual Design
- How can we design better interfaces?

Feedback (through anonymous form)



<https://forms.office.com/r/kEcdPkQCaE>

Feedback

- Email me directly:
m.al-banna@unsw.edu.au
- Talk to your mentor